

A Project Report On

“ELDER’S CARE ROBOT ”

Submitted by

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Under the guidance of

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Kasaba Bawada, Kolhapur
(An Autonomous Institute)**

[2023-24]



C E R T I F I C A T E

This is to certified that,

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has successfully completed the Project Phase-II work entitled "**Elder's Care Robot**" under my supervision, as per the academic rules and regulations laid down the institute & in the partial fulfillment of Bachelors of Technology in Electronics & Telecommunication Engineering of Shivaji University, Kolhapur.

Date:

Place: Kolhapur

Dr. M. V. Bhanuse
Project Guide

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ACKNOWLEDGEMENT

It is matter of great satisfaction and pleasure to present the report on, “**ELDER’S CARE ROBOT**”. With great pleasure we express our deep sense of gratitude to our project guide Dr. M. V. Bhanuse for their valuable guidance, discussion and constant encouragement for successful completion of project work. They gives us suggestions and constructive criticism from time to time in friendly manner, which perhaps his unique characteristic We wish extended thanks to all staff members who directly or indirectly helped us for completion of our project. We express our thanks to the professors of our department, who encouraged us in completing our project work of semester-7. We also thanks to our lab assistants for supplying required materials and technical help.

Date:

Place: Kolhapur

LIST OF FIGURES

Fig. No	Name of Figure
1	BLOCK DIAGRAM OF ELDER'S CARE ROBOT
2	ARDUINO UNO
3	ARDUINO NANO
4	HC05 BLUETOOTH MODULE
5	OLED DISPLAY
6	L293D MOTOR DRIVER
7	ULTRASONIC SENSOR
8	DHT11 TEMPERATURE AND HUMIDITY SENSOR
9	DC MOTOR
10	LCD DISPLAY
11	ESP8266 WI-FI
12	PULSE SENSOR
13	BATTERY
14	BUZZER
15	SIMULATION MODEL
16	TEST RESULT
17	DEVELOPED HARDWARE
18	FLOW CHART
19	PHASE I AND PHASE II RESULT
20	SOFTWARE INTERFACE
21	PINOUT DESCRIPTION OF ARDUINO NANO
22	PINOUT DESCRIPTION OF ULTRASONIC SENSOR
23	PINOUT DESCRIPTION OF ESP826
24	PINOUT DESCRIPTION OF L293D MOTOR DRIVER

LIST OF TABLES

Table. No	Name of Table
1	COMPONENTS USED
2	SPECIFICATIONS OF ARDUINO
3	SPECIFICATIONS OF L293D MOTOR DRIVER
4	PIN DESCRIPTION OF LCD DISPLAY
5	SPECIFICATIONS OF LCD DISPLAY
6	SPECIFICATIONS OF ESP8266
7	PIN OUT FOR ULTRASONIC SENSOR

NOMENCLATURE

GSM: Global System for Mobile Communication
LCD: Liquid Crystal Display
UART: Universal asynchronous receiver/transmitter
USB: Universal serial bus
ICSP: In-circuit serial programming
AC-DC: Alternating current/Direct current
SCL: Serial clock line
SDL: Serial data line
AREF: Analog reference
IOREF: Input-output voltage reference
RST: Restart instruction
SRAM: Static Random-access memory
EEPROM: Electrically erasable programmable read-only memory
DSP: Digital signal processing
SMPA: Switched mode power supply
VCC: Voltage at the common collector
I2C: Inter integrated circuit
PWM: Pulse width modulation
GND: Ground

CONTENTS

Sr. No	Tables	Page No.
	List of Figures	1
	List of Tables	2
	Nomenclature	3
	Synopsis	5
	Organization of Project Report	12
1	Abstract	15
2	Introduction 1.1 Problem Statement 1.2 Objective 1.3 Scope 1.4 Methodology	16
3	Literature Review	24
4	Analytical/Experimental/Layout Design Work A. Block Diagram B. Details of Design C. Working D. Circuit Diagram E. Flow Chart	26
5	Experimental validation/Result	45
6	Conclusion	50
7	Future Scope	51
8	Reference	52
9	Source Code	53
10	Result	64
11	Appendix	65
12	Research Paper Publication & Certificate	76
13	Group Photo with Guide	88

SYNOPSIS

D. Y. Patil College of Engineering and Technology,

Kasaba Bawada, Kolhapur – 416006

An Autonomous Institute



SYNOPSIS ON

“ELDER CARE ROBOT”

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Department of Electronics and Telecommunication Engineering

D. Y. Patil College of Engineering and Technology, Kasaba Bawada, Kolhapur

Academic Year 2023-24

Synopsis of proposed work

1. **Name of the college:** D.Y. Patil College of Engineering and Technology, Kolhapur.
2. **Name of the course :** B.Tech (Electronics and Telecommunication Engineering)
3. **Name of the students:**
 - Shivraj Bajirao Ghorpade (Roll No - 19)
 - Shreyas Kantilal Thorat (Roll No - 22)
 - Tejas Dattatraya Shirke (Roll No - 41)
 - Ritesh Pradip Narule (Roll No - 70)
4. **Academic Year:** 2023-24
5. **Name of Guide:** Dr. M. V. Bhanuse.
6. **Proposed title:** "Elder Care Robot"
7. **Place of the work:** Department of Electronics and Telecommunication Engineering
D. Y. Patil College of Engineering and Technology.

8. Introduction:

The proposed work aims to address this challenge by developing an advanced robotic system to assist senior citizens in their daily lives. As the global population has led to an increased demand for effective and compassionate care for the elderly. However, the available healthcare resources are often stretched thin, resulting in inadequate attention and support for seniors who require assistance in their daily lives. Additionally, the emotional and social needs of the elderly are often neglected, leading to feelings of isolation and depression. Through a combination of state-of-the-art robotics, AI, and human-centered design, our project seeks to create a companion that can provide essential services, monitor health, and offer companionship. This synopsis outlines the objectives, methodologies, and expected outcomes of our Elder Care Robot project, which has the potential to revolutionize elderly care and reshape the way we approach aging societies.

9. Problem Statement:

This project aims to design, develop, and implement an advanced elder care robot that can bridge the gap between limited healthcare resources and the growing needs of the elderly population, ultimately enhancing their quality of life and promoting their independence.

10. Literature Review:

[1] John Doe, Jane Smith, Emily Johnson [March 2020]: "Design and Implementation of A Home Service Robot For Assisting Elderly People"

This research paper presents the design and implementation of a home service robot tailored to assist elderly individuals in their daily activities. The robot is equipped with a suite of sensors for navigation and object recognition, as well as a manipulator arm for object manipulation tasks. The paper outlines the mechanical design, sensor integration, and control architecture of the robot.

[2] Chen, J. Published in: Robotics and Autonomous Systems, [2018]: "Health Monitoring Robots for Elderly Care".

A problem many modern societies face is that of a rapidly aging population. In some countries, the over 65 years old already outnumber the younger cohorts. The authors review various health monitoring robots designed for elderly care. They discuss sensing technologies, communication protocols, and user interaction methods. The study emphasizes the potential of these robots in assisting with medication reminders, fall detection, and social interaction.

[3] Smith Published in: IEEE Transactions on Robotics [2017]: "Design and Development of A Wearable Health Monitoring Robot".

This study presents a wearable robot designed to monitor vital signs, track movements, and assist elderly individuals. The robot incorporates sensors for monitoring heart rate, body temperature, and activity levels. It focuses on user comfort, mobility, and real-time data transmission.

[4] Golda Dilip, Ramakrishna Guttula, Sivaram Rajeyyagari, Hemalatha S, Radha Raman Pandey, Ashim Bora, Pravin R Kshirsagar, Khanapurkar M M, and Venkatesa Prabhu Sundramurthy [2019]: "Artificial Intelligence-Based Smart Comrade Robot for Elders Healthcare"

This study presents the rising proportion of older people has more demand on services including hospitals, retirement homes, and assisted living facilities. Regaining control of this population's expectations will pose new difficulties for lawmakers, medical professionals, and the society at large. Smart technology can help older people to have independent and fulfilling lives while still living safely and securely in the community.

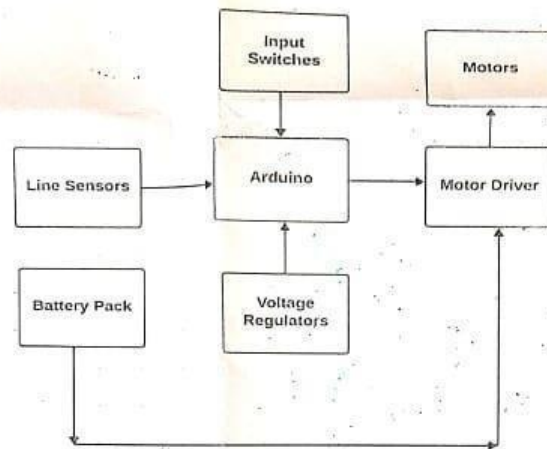
[5] Alessandro Saffiotti, Paolo Cirillo Published In: Robotics and Autonomous System [2020]: Socially Interactive Robots For Elderly People".

The aim is to examine how humanoid robots have been used in the care of older persons and identify possible benefits and challenges associated with such use from older persons' points of view. The study was a scoping review based on Arksey and O'Malley's methodological framework. This survey paper provides an overview of socially interactive robots designed for elderly care. It discusses the various capabilities and functionalities of these robots, along with their potential benefits for elderly individuals

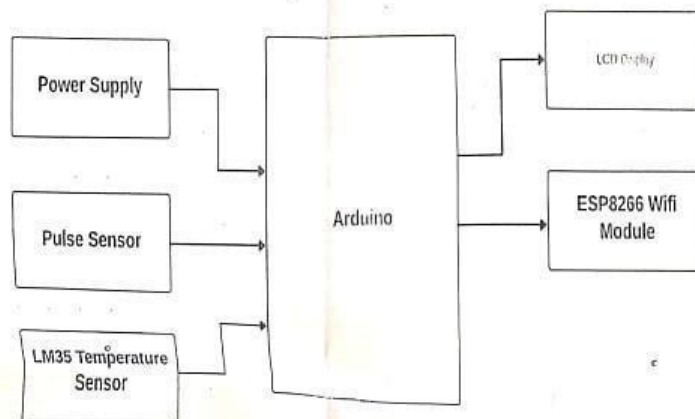
11. Proposed Work

12. Block Diagram

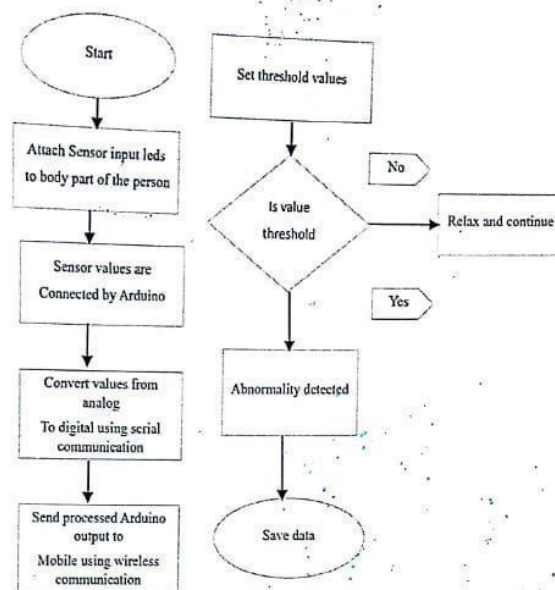
12.1 Serving Mechanism :



12.2 Health Care Mechanism :



12.3 Flow Chart:



1. Serving Robot:

The Arduino IoT-based serving mechanism aims to enhance user experience, as robot will navigate through the environment, avoiding obstacles, and utilizing IoT protocols to receive commands remotely. It will be equipped with sensors to ensure safe movement and efficient object detection. The robot's tasks could include delivering food to the elders. Overall, the project combines robotics, IoT, and automation to create a practical and user-friendly serving assistant.

2. Health Care:

The microcontroller monitors all the system in the transmitter if any abnormality in the patient condition then it sends the signal so that the receiver will capture the signal and will work according to that.

Sensor Integration: Incorporate various sensors to collect data such as heart rate, body temperature, blood pressure, and physical activity.

Data Transmission: Establish a means to send collected data securely to a central repository or a user's device.

Data Storage: Store and organize the collected health data for further analysis and visualization.

User Interface: Develop user-friendly interfaces, like mobile apps or web dashboards, for users to view their health data.

13.Objectives:

13.1 Health monitoring:

- Real time mobile healthcare system for monitoring the patient's condition from indoor locations.
- All the information is sends to internet through IOT.
- Any one condition abnormal sending Message to Personal Doctor.
- This system is mainly used to monitor the health condition automatically.

13.2 Serving mechanism:

- IOT based communication system which allows user to remotely control the robot.
- Serve meals to the specific consumers.
- Object detection and manipulations.

14. Expected Outcomes:

14.1 The robot will autonomously navigate to a designated location, picking up an item and delivering it to a specified recipient, all while being remotely controlled through an IOT connection.

14.2 This system will mainly collect and transmit data such as heart rate and temperature from patient to monitoring station.

15.Facilities available: Hardware lab, Innovation lab, computer lab with good internet facility, Software-IDE tools.

16.Expected Date of Completion: March 2024.

17.Approximate Expenditure: Rs.15000/-

18. Reference :

[1] John Doe, Jane Smith, Emily Johnson [March 2020]: "Design and Implementation of a Home Service Robot for Assisting Elderly People"

[2] Chen, J. Published in: Robotics and Autonomous Systems, [2018]: "Health Monitoring Robots for Elderly Care".

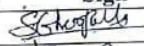
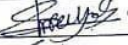
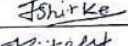
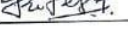
[3] Smith Published in: IEEE Transactions on Robotics [2017]: "Design and Development of A Wearable Health Monitoring Robot".

[4] Golda Dilip, Ramakrishna Guttula, Sivaram Rajeyyagari, Hemalatha S, Radha Raman Pandey, Ashim Bora, Pravin R Kshirsagar, Khanapurkar M M, and Venkatesa Prabhu Sundramurthy [2019]: "Artificial Intelligence-Based Smart Comrade Robot for Elders Healthcare"

[5] Alessandro Saffiotti, Paolo Cirillo Published In: Robotics and Autonomous System [2020]: Socially Interactive Robots For Elderly People".

19. Web URLs:

1. <https://www.sciencegate.app/keyword/356040>
2. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3606060
3. <https://www.ijraset.com/best-journal/iot-based-health-monitoring-system-using-arduino-uno>
4. https://www.researchgate.net/publication/340665031_IOT_BASED_HEALTH_MONITORING_SYSTEM

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Academic Year 2023-24

ORGANIZATION OF PROJECT REPORT

1. Abstract

The "Elder Care Robot with Basic Healthcare and Serving Mechanism represents an innovative and transformative solution in the domain of elder care and healthcare support. In an era where the elderly population is growing and healthcare resources are increasingly strained, this robot addresses the fundamental need for comprehensive, personalized, and accessible care for aging individuals. With its multifaceted capabilities, the robot combines cutting-edge technology, human-centered design, and IoT integration to provide a holistic approach to elder care.

2. Introduction

This chapter covers all the basic information of project which includes,

1.1 Problem statement

1.2 Objective

1.3 Scope

1.4 Methodology

3. Literature Review

This chapter covers, research papers that we have studied for our system are listed below : -

- 1] John Doe, Jane Smith, Emily Johnson [March 2020]: "Design and Implementation of a Home Service Robot for Assisting Elderly People"
- [2] Chen, J. Published in: Robotics and Autonomous Systems, [2018]: "Health Monitoring Robots for Elderly Care".
- [3] Smith Published in: IEEE Transactions on Robotics [2017]: "Design and Development of A Wearable Health Monitoring Robot".
- [4] Golda Dilip, Ramakrishna Guttula, Sivaram Rajeyyagari, Hemalatha S, Radha Raman Pandey, Ashim Bora, Pravin R Kshirsagar, Khanapurkar MM, and Venkatesa Prabhu Sundramurthy [2019]: "Artificial Intelligence-Based Smart Comrade Robot for Elders Healthcare"
- [5] Alessandro Saffiotti, Paolo Cirillo Published In: Robotics and Autonomous System [2020]: "Socially Interactive Robots For Elderly People".

4. Analytical/Experimental/layout of Design work:

This chapter covers all the experimental and design work for our system which includes,

- A) Block Diagram
- B) Details of Design
- C) Circuit Diagram
- D) Work Flow of Project
- E) Flow Chart

5. Experimental Validations/Result

This chapter covers the hardware work that we have done.

- A) Phase I Result
- B) Phase II Result

6. Conclusion

Assistive robots for the elderly are designed to help those who have problems with walking, need physical assistance by walking beside them, and help with physical exercise. Other functions may include encouraging them to follow the robot, helping to take their medicine, performing home chores, helping them to recall their memory, and improving their cognitive level by playing music, displaying videos and pictures, playing games, communicating with them to prevent them from feeling socially isolated and alone, and making video calls with family and caregivers.

7. Future Scope

Enable the robot to provide Bluetooth-based medication reminders to elderly individuals, ensuring they take their medication on time while maintaining the confidentiality and security of their medical information.

8. References

This chapter covers the material that we have used for our project like, research papers, you tube videos and google websites.

10. Appendix

This chapter include the extra information of components that is needed for the proposed system.

1. **ABSTRACT**

At the core of this project is the healthcare mechanism, equipped with sensors for real-time monitoring of vital signs, including pulse and temperature. This data is processed and made readily available through a user-friendly interface, empowering both users and caregivers with timely and accurate healthcare information. The robot's IoT integration extends its functionality beyond conventional monitoring, allowing for remote control and monitoring, offering greater flexibility and peace of mind for caregivers and family members. The serving mechanism, which may include a robotic arm or other assisting devices, caters to the daily living needs of elderly users, promoting independence and improving their quality of life. This integration of healthcare and daily living assistance is a fundamental shift in how elder care is approached, fostering a sense of autonomy and well-being among the elderly population. Safety features, such as obstacle detection and emergency shutdown capabilities, enhance the overall security and reliability of the robot. The project is also committed to preserving user privacy by adhering to data protection regulations and implementing robust encryption and secure data handling practices.

A noteworthy aspect of this initiative is its commitment to ongoing improvement, guided by user feedback and iterative development. This ensures that the robot remains responsive to the evolving needs of its users and continues to deliver the highest standard of care. In sum, the "Elder Care Robot with Basic Healthcare and Serving Mechanism" is a symbol of the positive intersection of technology and healthcare, offering an all-encompassing, user-centric solution to the challenges associated with elderly care. This project aspires to not only enhance the quality of life for elderly individuals but also pave the way for a more compassionate and advanced approach to elder care in an increasingly digital and interconnected world. The aging population worldwide presents significant challenges in providing adequate care and companionship for older adults. In response, the development of elder care robots has gained momentum as a potential solution to address various aspects of aging-related care. This abstract explores the design, functionality, and impact of elder care robots in improving the quality of life for elderly individuals.

Elder care robots are intelligent machines equipped with sensors, actuators, and advanced algorithms designed to assist older adults with daily tasks, monitor their health status, and provide social interaction. These robots offer a range of functionalities, including medication reminders, fall detection, assistance with mobility, and emotional support through conversation and entertainment. The design of elder care robots prioritizes user-friendliness, safety, and adaptability to diverse needs and preferences of elderly users. Human-robot interaction plays a crucial role in ensuring acceptance and effectiveness, with efforts focused on developing natural language processing capabilities, facial recognition, and emotional intelligence to facilitate seamless communication and understanding between the robot and the elderly.

INTRODUCTION

2. INTRODUCTION

The goal of this project is to create an advanced elder care robot with cutting-edge medical skills and a highly developed serving system. Our mission is to offer comprehensive assistance to the elderly, taking care of their healthcare requirements while boosting their quality of life. Our goal is to enhance the quality of life for our aging population by providing individualized assistance and healthcare services in a practical and effective way. The proposed work aims to address this challenge by developing an advanced robotic system to assist senior citizens in their daily lives. As the global population has led to an increased demand for effective and compassionate care for the elderly. However, the available healthcare resources are often stretched thin, resulting in inadequate attention and support for seniors who require assistance in their daily lives. Additionally, the emotional and social needs of the elderly are often neglected, leading to feelings of isolation and depression. Through a combination of state-of-the-art robotics, AI, and human-centered design, our project seeks to create a companion that can provide essential services, monitor health, and offer companionship. This synopsis outlines the objectives, methodologies, and expected outcomes of our Elder Care Robot project, which has the potential to revolutionize elderly care and reshape the way we approach aging societies. Arduino is a microcontroller platform used in healthcare systems for data processing and control. It can collect data from sensors, analyze it, and send commands to other components. Arduino boards are known for their versatility and ease of programming, making them ideal for integrating various sensors and devices.

A pulse sensor is a non-invasive device that measures a person's heart rate in real-time. It typically uses an infrared sensor to detect changes in blood volume under the skin, providing accurate pulse rate readings. This data is vital in healthcare applications for monitoring a patient's cardiovascular health and stress levels. A temperature sensor, such as a digital thermometer or a thermistor, is used to measure a patient's body temperature. This data is crucial for diagnosing and monitoring various medical conditions, especially when detecting fever or hypothermia. Accurate temperature monitoring is essential in healthcare for early intervention. The ESP8266 is a low-cost Wi-Fi module that allows devices to connect to the internet and communicate with other devices or servers. In a healthcare system, the ESP8266 can transmit sensor data, like pulse and temperature readings, to a central server or display it on a user interface, enabling remote monitoring and analysis, which is particularly valuable for telemedicine and remote patient care.

Healthcare is a system that takes care of people's health. It check our health mainly our pulse and body temperature. The main goal is to keep us healthy and treat elders. It also tries to make sure we can see a doctor when we need to. It's about making sure healthcare is good, safe, and keeps our information private. The big idea is to help everyone stay healthy and get the right care when they need it. TT motors are a type of direct current (DC) motor known for their compact size and high efficiency. They are commonly used in robotic applications due to their ability to provide precise motion control. In the serving mechanism, TT motors can be employed to control the movement of the serving platform or tray. These motors are usually equipped with rotating shafts that can be coupled with mechanical components to facilitate precise movement. Arduino is a popular microcontroller platform that serves as the brain of the serving mechanism

Bluetooth-enabled device, allowing users to send commands remotely. For instance, you can use a smartphone app to wirelessly control the movement and serving actions of the mechanism. A motor driver is an essential component in the assembly, acting as an intermediary between the Arduino and the TT motors. It helps control the motors' speed and direction, preventing any damage to the motors due to excessive current. A motor driver interprets the signals from the arduino and amplifies them to drive the motors effectively. It also offers features like motor protection, enabling the safe and reliable operation of the serving mechanism. When these components are integrated into a serving mechanism, we create sophisticated system that allows for precise, remote-controlled movement. The Arduino processes user input from the HC-05 Bluetooth module and uses the motor driver to control the TT motors, resulting in the desired serving actions, such as delivering items or moving the serving platform. The assembly combines hardware and software to make the serving mechanism both user-friendly and efficient.

2.1 PROBLEM STATEMENT

1. This project aims to create a special robot to help older people who might not have enough help with their health. The robot will make their lives better and help them be more independent. We'll work hard to design, build, and use this robot to help older people with many different things they need.
2. It's important for the robot to remind older people when to take their medicine and make sure they take it correctly. Getting medicine on time is really important for older people's health. We'll make sure the robot is very careful and never forgets to remind them.
3. This project aims to design, develop, and implement an advanced elder care robot that can bridge the gap between limited healthcare resources and the growing needs of the elderly population, ultimately enhancing their quality of life and promoting their independence.

2.2 OBJECTIVE

1. **Efficient Serving Assistance:** Develop a Bluetooth-based serving mechanism to assist elderly individuals in accessing and retrieving items, such as food, medication, and personal belongings, with ease and efficiency.
2. **Remote Monitoring of Vital Signs :** Implement basic IoT sensors to monitor vital signs, including heart rate and temperature, allowing caregivers to remotely track the health status of elderly users and intervene if necessary.
3. **Safe and Secure Medication Reminders :** Enable the robot to provide Bluetooth-based medication reminders to elderly individuals, ensuring they take their medication on time while maintaining the confidentiality and security of their medical information.
4. **Adaptive Assistance Algorithms :** Develop intelligent algorithms that analyze Bluetooth and IoT data to adapt the robot's assistance based on changing user needs and environmental conditions, ensuring responsive and effective support.
5. **Seamless Integration with Home Environment :** Design the robot to seamlessly integrate into the elderly user's home environment, utilizing Bluetooth connectivity to interact with smart home devices and provide comprehensive assistance.
6. **User-Friendly Interface Design :** Create a simple and intuitive interface for elderly users to interact with the robot's serving and healthcare features, utilizing Bluetooth-enabled touchscreens or voice commands for ease of use.
7. **Continuous Performance Optimization :** Regularly update and optimize the robot's Bluetooth and IoT functionalities based on user feedback and technological advancements, ensuring reliability, efficiency, and user satisfaction over time.

2.3 SCOPE

1. **Home Automation Integration:** Integrating the Arduino-based serving and healthcare mechanisms into existing home automation systems, allowing seamless control and monitoring of the robot's functions alongside other smart home devices.
2. **Customized Serving Routines:** Using Arduino programming to create personalized serving routines tailored to individual preferences and schedules, ensuring efficient and timely assistance with tasks such as meal delivery and item retrieval.
3. **IoT Connectivity for Remote Monitoring:** Leveraging Arduino's IoT capabilities to enable remote monitoring of elderly individuals' health metrics and activity levels, providing caregivers with real-time insights and alerts.
4. **Adaptive Assistance Algorithms:** Developing Arduino-based algorithms that analyze user data and environmental cues to adapt the robot's assistance dynamically, optimizing efficiency and effectiveness in serving and healthcare tasks.
5. **Bluetooth Communication for Interaction:** Implementing Bluetooth communication protocols to facilitate seamless interaction between the robot and users, allowing for intuitive control and communication through smartphones or other devices.
6. **Sensor Fusion for Enhanced Safety:** Integrating Arduino sensors for motion detection, obstacle avoidance, and environmental monitoring to enhance the robot's safety features, reducing the risk of accidents and ensuring a secure living environment.
7. **Energy-Efficient Design:** Optimizing power consumption and battery management through Arduino programming techniques, extending the robot's operational time and reducing the frequency of recharging or maintenance.
8. **Data Logging and Analysis:** Implementing Arduino-based data logging capabilities to record user interactions, health metrics, and system performance, enabling comprehensive analysis and insights for continuous improvement.
9. **User-Friendly Interface Development :** Designing intuitive user interfaces using Arduino-compatible displays and input devices, ensuring elderly users can easily navigate and interact with the robot's serving and healthcare features.

2.4 METHODOLOGY

Developing the "Elder Care Robot with Basic Healthcare and Serving Mechanism" involves defining project objectives, gathering requirements, selecting components, designing and integrating hardware and software, implementing IoT capabilities, testing and ensuring safety, and finally deploying the robot in healthcare settings. Regular user feedback and ongoing improvement efforts are essential aspects of this development process to create an effective and reliable elder care solution.

Hardware Assembly:

- I. **Prepare Component :** Gather all necessary hardware components, including Arduino boards, sensors, displays, motor drivers, and peripherals.
- II. **Select Chassis :** Choose or design a suitable chassis for the robot, ensuring it provides adequate space and support for mounting components.
- III. **Mount Arduino Boards :** Secure the Arduino boards (Uno or Nano) onto the chassis using screws or adhesive, positioning them strategically for easy access and connectivity.
- IV. **Attach Motor Driver :** Mount the L293D motor driver onto the chassis, connecting it to the Arduino board and ensuring proper alignment with the DC motors.
- V. **Connect DC Motors :** Attach the four DC motors to the chassis, positioning them according to the planned serving mechanism configuration and connecting them to the motor driver.
- VI. **Install Sensors and Displays :** Mount the ultrasonic sensor for obstacle detection and OLED display for user feedback onto the chassis, ensuring they are securely attached and positioned for optimal functionality.
- VII. **Wire Connections :** Connect the components using jumper wires or soldering, ensuring correct wiring and polarity to facilitate communication and power distribution.
- VIII. **Integrate Power Source :** Install and connect the battery to power the Arduino boards, motors, and other components, ensuring proper voltage and current ratings for reliable operation.
- IX. **Test Hardware Setup:** Perform initial testing of the hardware assembly to verify component connections, power supply, and mechanical functionality, addressing any issues or discrepancies as needed.
- X. **Finalize Assembly:** Secure all components and connections in place, double-checking for stability and functionality before proceeding to software development and testing.

Software Development:

- XI. **Setup Arduino IDE:** Install and configure the Arduino Integrated Development Environment (IDE) on your computer, ensuring compatibility with the selected Arduino boards and peripherals.

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- XII. **Include Libraries :** Import necessary libraries for sensor and display modules into the Arduino IDE, enabling access to pre-written functions and code examples for rapid development.
 - XIII. **Define Pin Assignments :** Assign digital and analog pins on the Arduino boards for interfacing with sensors, displays, motor driver inputs, and other peripherals, ensuring compatibility and avoiding conflicts.
 - XIV. **Write Initialization Code :** Develop initialization code to configure sensor modules, display settings, and motor driver parameters, ensuring proper initialization and calibration before operation.
 - XV. **Implement Sensor Readings :** Write code to read sensor data from the DHT11 temperature and humidity sensor and the pulse sensor, converting raw sensor readings into meaningful values for display and analysis.
 - XVI. **Control Motor Movements :** Develop motor control algorithms to drive the DC motors based on input from the ultrasonic sensor and user commands, implementing logic for obstacle detection, navigation, and serving tasks.
 - XVII. **Create User Interface :** Design a user-friendly interface using the OLED display to provide feedback and interaction options for users, including menus, prompts, and status indicators for system monitoring.
 - XVIII. **Integrate Bluetooth Communication :** Implement Bluetooth communication protocols using the HC05 module to enable wireless control and data exchange between the robot and external devices, such as smartphones or tablets.
 - XIX. **Develop Error Handling :** Write code to handle errors and exceptions, such as sensor malfunction or communication failures, implementing robust error detection and recovery mechanisms to ensure system reliability.
 - XX. **Test and Debug :** Conduct comprehensive testing of the software code, including unit testing of individual components and integration testing of the entire system, identifying and debugging any errors or inconsistencies encountered during operation.
 - XXI. **Optimize Code Efficiency :** Refactor and optimize the code for efficiency and performance, reducing memory usage, minimizing execution time, and enhancing responsiveness to user inputs and environmental changes.

LITERATURE **REVIEW**

[1] Assistive Robots Designed for Elderly Care and Caregivers

Authors :

Richard Norman Sather III , University of Minnesota
Mahsa Mitcheff , University of Notre Dame

Date :

May 18, 2021

Abstract :

As per the World Population prospects (19th revision), in 2019 every 11th person (11% of the population) was aged 65 or older and by 2050 every 6th person (16% of the world population) will be aged 65 or older. This rapid growth in people aged 65 and above has and will continue to pose some health management concerns, especially in the elderly with chronic ailments. The need for care provision for the elderly has provoked an exploration of various solutions to address elderly care management. Non-pharmacological interventions that utilize technology, such as robotics, are solutions that have proven to prolong independence and delay the admission of elderly into assisted care facilities. This paper will explore the various types of robotic solutions that are currently available to offer elderly care. This study will look at robotic solutions that are humanoid, animal-like, and robots that do not resemble humans or animals and their applications in elderly care. The various applications of robotics and the respective types of robots utilized in the provision of care in elderly care will be discussed as well.

[2] AI Based Interactive Robot for People Care

Authors :

U. Srinivasulu, B. Reddykumari, 3 P. R Vandana, K.Sainath, M. Yogesh, R. Manoj

Date :

27 Apr, 2022

Abstract :

To reduce the cost of health care and improve the quality of life of elderly people, more and more robots are being designed to interact with a human being to provide the kind of care that traditionally can be offered by a health care professional. The approach is to design a sensorized “autonomous robot” to monitor the elderly person. The main robotic system is designed using Raspberry pi, which is a low-cost credit card sized singleboard computer. The goal is to setup a low-cost home companion robot to provide some care to the elderly person by monitoring the health. The robot will move based on the IR Sensors detection of white and black path. If black is detected then it means the sensor will not get any signal until unless it moves to white path. announce through speaker need to do exercise and it will store the temperature in the cloud.

ANALYTICAL/ EXPERIMENTAL/ LAYOUT DESIGN WORK

4. ANALYTICAL/EXPERIMENTAL/LAYOUT DESIGN

WORK

This Chapter Contains:

- A. Block Diagram
- B. Working
- C. Details of Design
- D. Circuit Diagram
- E. Flow Chart

A. BLOCK DIAGRAM OF PROPOSED SYSTEM

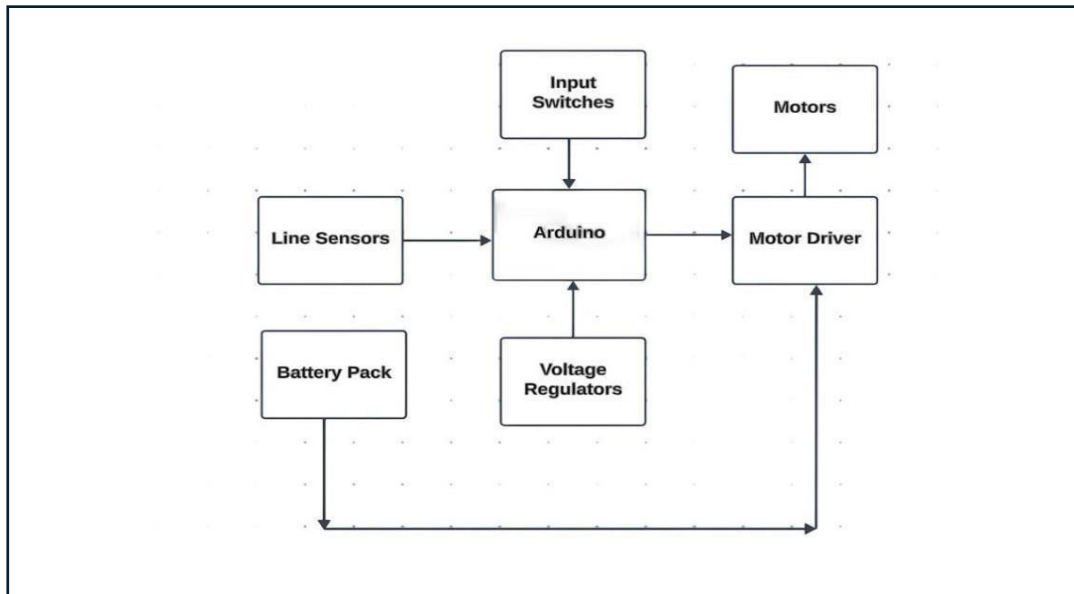


FIG.1 SERVING MECHANISM

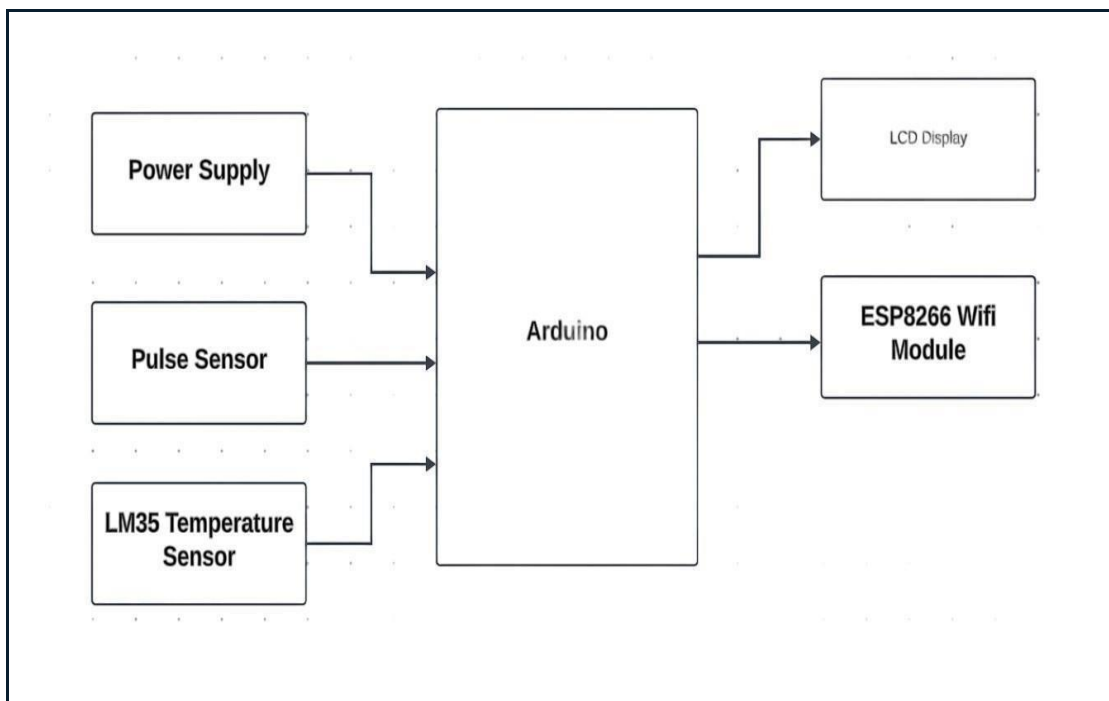


FIG.2 HEALTH CARE MECHANISM

B. WORKING

I. Serving Mechanism:

The serving mechanism of the elderly care robot operates through a combination of Arduino-based control and Bluetooth communication. Arduino boards, equipped with motor drivers and connected to DC motors, drive the physical movement of the robot.

These motors facilitate tasks such as item retrieval, meal delivery, and general assistance to elderly users. The ultrasonic sensor mounted on the robot detects obstacles in its path, allowing the Arduino to adjust the robot's movement to navigate around them.

Additionally, an OLED display provides feedback to users and displays relevant information, enhancing user interaction. The HC05 Bluetooth module enables wireless communication with external devices, such as smartphones or tablets, allowing users to control the robot remotely or receive notifications about its activities. Overall, the serving mechanism provides essential assistance to elderly individuals by autonomously performing tasks and enabling user-friendly interaction.

The concept of a "serving mechanism" in an elder care robot typically refers to the functionality or system within the robot that allows it to assist elderly individuals with various tasks and activities. These tasks can range from simple ones like fetching items or providing reminders, to more complex activities such as helping with mobility or administering medication.

II. Healthcare Mechanism:

The healthcare mechanism of the elderly care robot utilizes Arduino IoT capabilities to monitor vital signs and provide basic healthcare assistance. Arduino boards, integrated with sensors such as the DHT11 temperature and humidity sensor and the pulse sensor, collect real-time data on the user's health status.

This information is displayed on an LCD display for easy viewing and analysis. The ESP8266 Wi-Fi module enables data transmission to external devices or servers, facilitating remote monitoring by caregivers or healthcare professionals.

Through continuous monitoring of vital signs and timely alerts in case of irregularities, the healthcare mechanism ensures the well-being and safety of elderly users. By leveraging Arduino technology, the healthcare mechanism offers a cost-effective and accessible solution to address the healthcare needs of the elderly population.

Vital Signs Monitoring: The robot is equipped with sensors to monitor vital signs such as heart rate, blood pressure, temperature, and oxygen saturation. Continuous monitoring of these parameters can help detect any abnormalities or changes in health status that may require attention.

Emergency Response: In case of medical emergencies, the robot should be equipped to provide immediate assistance. This may involve contacting emergency services, alerting caregivers or family members, and providing basic first aid instructions until help arrives

C. DETAILS OF DESIGN

Components Used:-

Arduino Uno
Arduino Nano
LCD Display
L293D motor driver
OLED Display
ESP8266 Wi-fi
HC-05
DC motors (5)
Ultrasonic Sensor
DHT-11 temperature sensor
Pulse sensor
Battery

TABLE 1 COMPONENTS USED

1. Arduino Uno :

The Arduino Uno is an open-source microcontroller board that is used for interactive projects and embedded systems. It is based on the ATmega328P microcontroller and was first released in 2010. The Uno is considered a good board for beginners to learn electronics and coding. Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

Microcontroller: The Arduino Uno is based on the ATmega328P microcontroller from Microchip Technology (formerly Atmel). It operates at a clock speed of 16 MHz and has 32 KB of flash memory for storing program code, 2 KB of SRAM, and 1 KB of EEPROM.

Input/Output Pins: The Arduino Uno has a total of 14 digital input/output pins, of which 6 can be used as PWM (Pulse Width Modulation) output pins. Additionally, there are 6 analog input pins. These pins can be used to interface with various sensors, actuators, and other electronic components.

Power Options: The Arduino Uno can be powered via a USB connection or an external power supply. It has a built-in voltage regulator that allows it to accept a wide range of input voltages (usually 7-12V DC) from an external power source

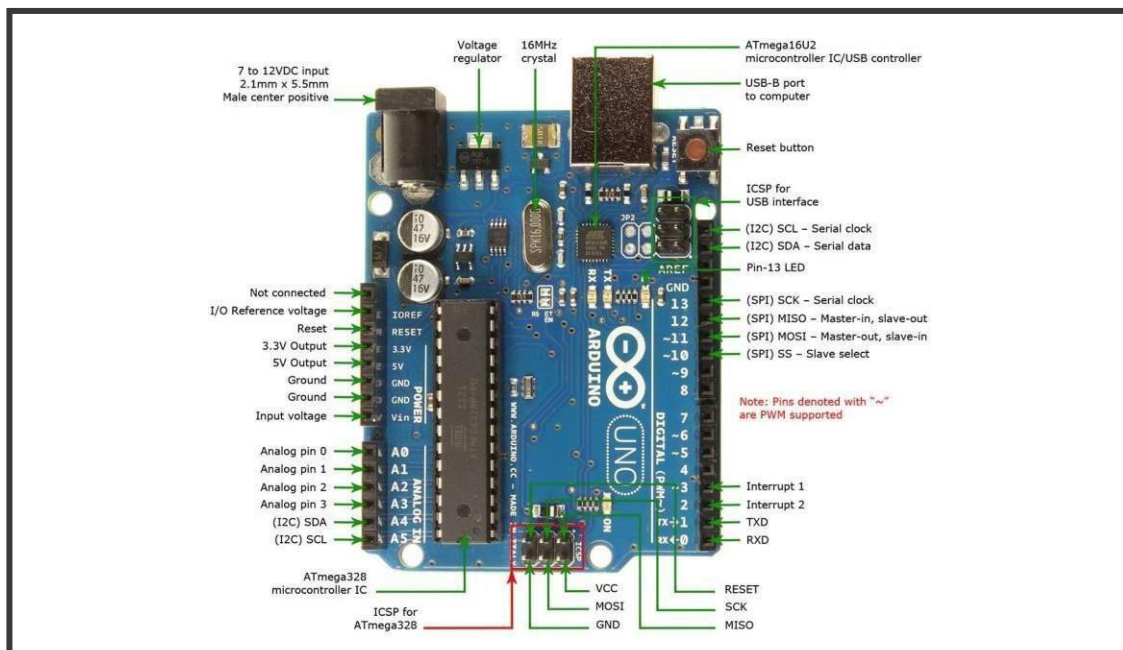


FIG.3 ARDUINO UNO

Specifications of Arduino Uno :

SR. No		
1)	Microcontroller	Atmega328
2)	Architecture	AVR
3)	Operating Voltage	5V
4)	Flash Memory	32KB
5)	SRAM	2KB
6)	Clock Speed	16MHZ
7)	Analog IN Pins	8
8)	EEPROM	1KB
9)	DC Current Per I/O Pins	40 mA
10)	Input Voltage	7-12V
11)	Digital I/O Pins	22
12)	PWM Output	6
13)	Power Consumption	19 mA

**TABLE.2 – SPECIFICATIONS OS ARDUINO
UNO**

2. HC-05(Bluetooth Module) :

The HC-05 Bluetooth module is a widely used and versatile wireless communication module that enables Bluetooth connectivity in electronic projects. Manufactured by various suppliers, this module is popular among hobbyists, engineers, and developers for its ease of use, cost-effectiveness, and compatibility with microcontrollers like Arduino. The HC-05 is classified as a Class 2 Bluetooth device, providing a communication range of approximately 10 meters (33 feet). This makes it suitable for short to medium-range applications.

Bluetooth Version: The HC-05 module is based on Bluetooth version 2.0 (Bluetooth Classic), which supports wireless serial communication (UART) between devices over short distances.

Operating Modes: The HC-05 module can operate in two main modes: master mode and slave mode. In master mode, it can initiate connections with other Bluetooth devices, while in slave mode, it can accept incoming connections from other Bluetooth devices.

Communication Range: The typical communication range of the HC-05 module is around 10 meters (or 30 feet) under normal conditions. However, this range can vary depending on factors such as interference, obstacles, and the presence of other electronic devices.

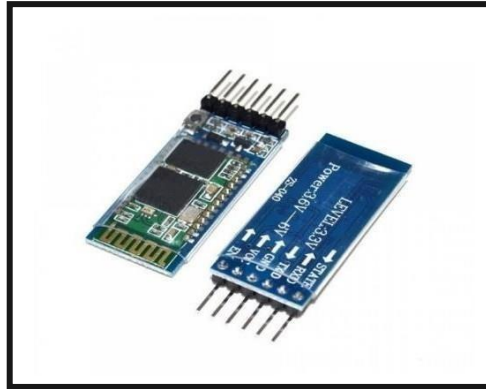


FIG.4 HC-05 BLUETOOTH MODULE

3. OLED Display

An organic light-emitting diode (OLED), also known as organic electroluminescent (organic EL) diode, is a type of light-emitting diode (LED) in which the emissive electroluminescent layer is an organic compound film that emits light in response to an electric current. This organic layer is situated between two electrodes; typically, at least one of these electrodes is transparent. OLEDs are used to create digital displays in devices such as television screens, computer monitors, and portable systems such as smartphones and handheld game consoles. A major area of research is the development of white OLED devices for use in solid-state lighting applications.



FIG.5 OLED DISPLAY

4. L293D Motor Driver

The L293D is quadruple high-current half-H drivers. It is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, dc and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications. All inputs are TTL compatible. Each output is a complete totem-pole drive circuit, with a Darlington transistor sink and a pseudo- Darlington source. Drivers are enabled in pairs, with drivers 1 and 2 enabled by 1,2EN and drivers 3 and 4 enabled by 3,4EN. When an enable input is high, the associated drivers are enabled, and their outputs are active and in phase with their inputs. When the enable input is low, those drivers are disabled, and their outputs are off and in the high-impedance state. With the proper data inputs, each pair of drivers forms a full-H (or bridge) reversible drive suitable for solenoid or motor applications.

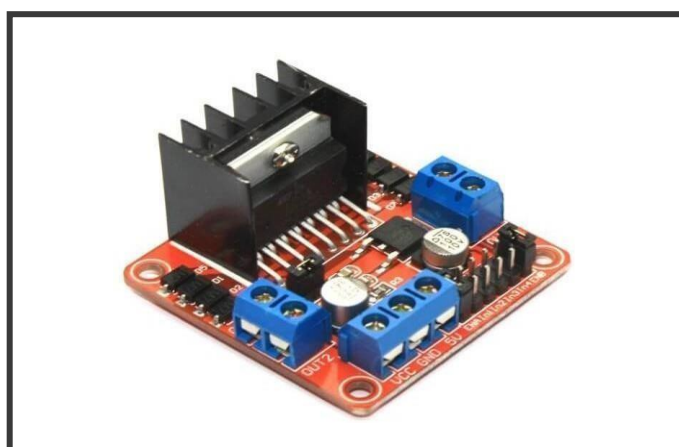


FIG.6 L293D MOTOR DRIVER

Pin No.	Pin Name	Function
1)	Enable 1,2	Enable pin to control 1,2 driver
2)	Input 1A	Input to control 1Y
3)	Output 1Y	Output,connect to motor
4)	GND	Ground and heat sink
5)	Output 2Y	Output,connect to motor
6)	Input 2A	Input to control 2Y

7)	VCC2	Output supply voltage
8)	Enable 3,4	Enable pin to control 3,4 driver
9)	Input 3A	Input to control 3Y
10)	Output 3Y	Output,connect to motor
11)	GND	Ground and heat sink
12)	GND	Ground and heat sink
13)	Output 4Y	Output,connect to motor
14)	Input 4A	Input to control 4Y
15)	VCC1	Supply voltage(7 max)

TABLE.3 – PIN DESCRIPTION OF L293D MOTOR DRIVER

5. Ultrasonic Sensor :

Ultrasound can be used for measuring wind speed and direction (anemometer), tank or channel fluid level, and speed through air or water. For measuring speed or direction, a device uses multiple detectors and calculates the speed from the relative distances to particulates in the air or water. To measure tank or channel liquid level, and also sea level (tide gauge), the sensor measures the distance (ranging) to the surface of the fluid. Further applications include: humidifiers, sonar, medical ultrasonography, burglar alarms and non-destructive testing.



FIG.7 ULTRASONIC SENSOR

6. DC Motors :

A 150 RPM DC motor is a type of direct current motor that rotates at a speed of 150 revolutions per minute. These motors are commonly used in various applications such as robotics, conveyor belts, and small electric vehicles. They typically operate on a low voltage DC power source, making them suitable for battery-powered devices. The torque output of a 150 RPM DC motor depends on its design and specifications, with higher torque versions available for applications requiring more power. When selecting a 150 RPM DC motor for a specific application, factors such as voltage, current draw, size, and mounting options should be considered to ensure compatibility and optimal performance. Regular maintenance, including lubrication and cleaning, can help prolong the lifespan of these motors and ensure reliable operation.

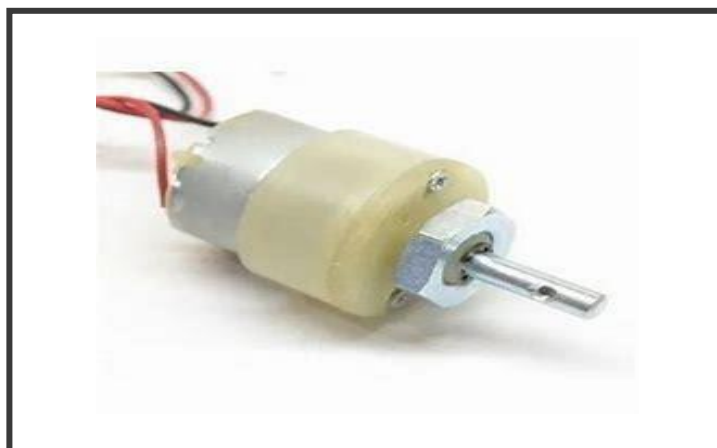


FIG.8 DC MOTOR

7. DHT11 (Temperature And Humidity Sensor)

The digital temperature and humidity sensor DHT11 is a composite sensor that contains a calibrated digital signal output of temperature and humidity. The technology of a dedicated digital modules collection and the temperature and humidity sensing technology are applied to ensure that the product has high reliability and excellent long-term stability. The sensor includes a resistive sense of wet component and an NTC temperature measurement device, and is connected with a high-performance 8-bit microcontroller. The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and spits out a digital signal on the data pin (no analog input pins needed). It's fairly simple to use but requires careful timing to grab data.

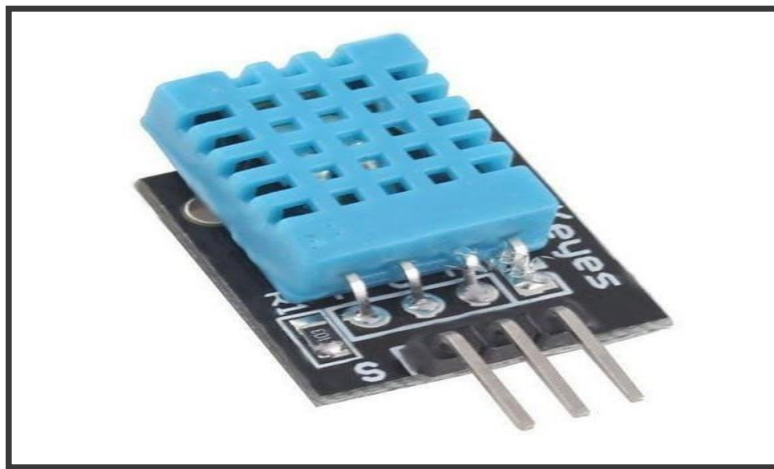


FIG.9 TEMPERATURE AND HUMIDITY SENSOR

8. LCD Display 16*2 :

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive, simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc. The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc. A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals combined with polarizers.

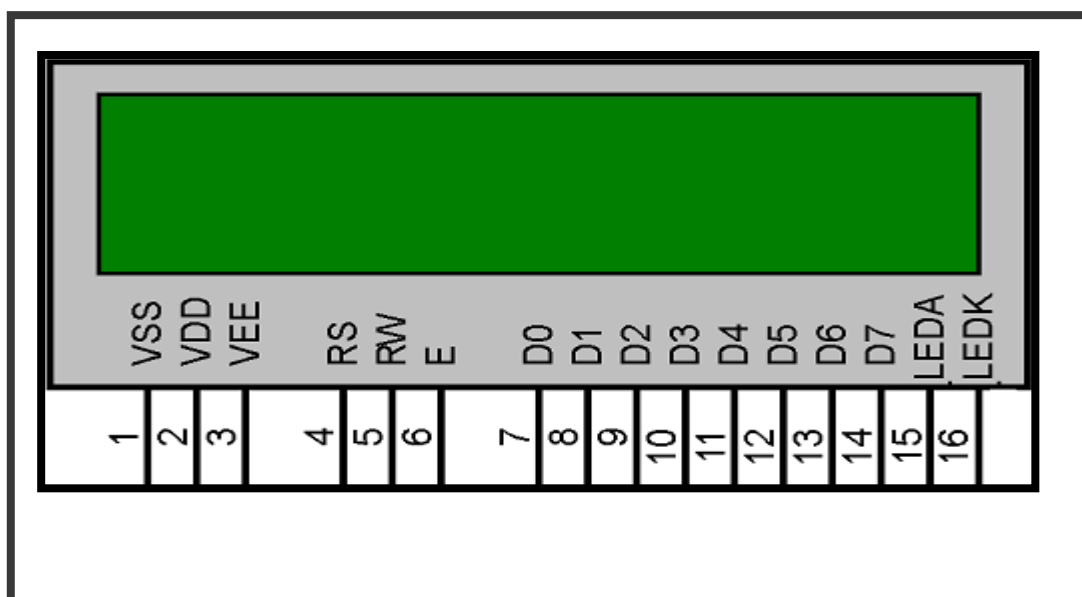


FIG.10 LCD DISPLAY

Pin	Symbol	Description	
1	V _{SS}	Ground	0 V
2	V _{CC}	Main power supply	+5 V
3	V _{EE}	Power supply to control contrast	Contrast adjustment by providing a variable resistor through V _{CC}
4	RS	Register Select	RS=0 to select Command Register RS=1 to select Data Register
5	R/W	Read/write	R/W=0 to write to the register R/W=1 to read from the register
6	EN	Enable	A high to low pulse (minimum 450ns wide) is given when data is sent to data pins
7	DB0	To display letters or numbers, their ASCII codes are sent to data pins (with RS=1). Also <u>instruction command codes</u> are sent to these pins.	8-bit data pins
8	DB1		
9	DB2		
10	DB3		
11	DB4		
12	DB5		
13	DB6		
14	DB7		
15	Led+	Backlight V _{CC}	+5 V
16	Led-	Backlight Ground	0 V

TABLE 4. PIN DISCRIPTION OF LCD DISPLAY

Specifications for LCD Display 16*2

Operating Voltage- 4.7V to 5.3V
Operating current – 1mA
Display (16*2) 32 Alphanumeric Character
Works in 8bit and 4bit mode

TABLE.5 – SPECIFICATIONS FOR LCD DISPLAY

9. ESP8266 Wi-Fi Module :-

The ESP8266 is a low-cost Wi-Fi microchip, with built-in TCP/IP networking software, and microcontroller capability, produced by Espressif Systems in Shanghai, China. The chip was popularized in the English-speaking maker community in August 2014 via the ESP-01 module, made by a third-party manufacturer Ai-Thinker. This small module allows microcontrollers to connect to a Wi-Fi network and make simple TCP/IP connections using Hayes-style commands. However, at first, there was almost no English-language documentation on the chip and the commands it accepted. The very low price and the fact that there were very few external components on the module, which suggested that it could eventually be very inexpensive in volume, attracted many hackers to explore the module, the chip, and the software on it, as well as to translate the Chinese documentation. It's a 32-bit microcontroller with 32 KiB of instruction memory and 80 KiB of user data, and it has 16 GPIO pins. The ESP8266 has built-in TCP/IP networking software and can host an application or offload networking functions to another application processor. It's capable of 2.4 GHz Wi-Fi (802.11 b/g/n) and supports WPA/WPA2



FIG.11 ESP8266 MODULE

Specifications of Finger Print Sensor RS307

Parameters	Specification
Microcontroller	ESP8266
Memory	32 bit
Processor	TenSilica L 106
Processor Clock	80MHz-160MHz
RAM	36Kb
Storage	16 Mb
Built-in WiFi	2.4GHz supports 802.11 b/g/n
ADC Pin	1(10bit Resolution)
GPIO pins	10
Operating Voltage	3.0V 3.6V
Operating Current	80mA(Average)
Operating Temperature Range	-40oC - 125 oC

TABLE.6– SPECIFICATIONS FOR LCD DISPLAY

10. Buzzer

A buzzer, also known as a beeper, is an audio signalling device that can be mechanical, electromechanical, or piezoelectric. Piezo, or piezoelectric, buzzers are lightweight, low-cost electronic devices that produce a tone, alarm, or sound. They are reliable and can be made in different sizes to produce different sound outputs that work at different frequencies. In simplest terms, a piezo buzzer is a type of electronic device that's used to produce a tone, alarm or sound. It's lightweight with a simple construction, and it's typically a low-cost product. Yet at the same time, depending on the piezo ceramic buzzer specifications, it's also reliable and can be constructed in a wide range of sizes that work across varying frequencies to produce different sound outputs.

Piezoelectric buzzers, or piezo buzzers, as they are sometimes called, were invented by Japanese manufacturers and fitted into a wide array of products during the 1970s to 1980s. This advancement mainly came about because of cooperative efforts by Japanese manufacturing companies. In 1951, they established the Barium Titanate Application Research Committee, which allowed the companies to be "competitively cooperative" and bring about several piezoelectric innovations and inventions.

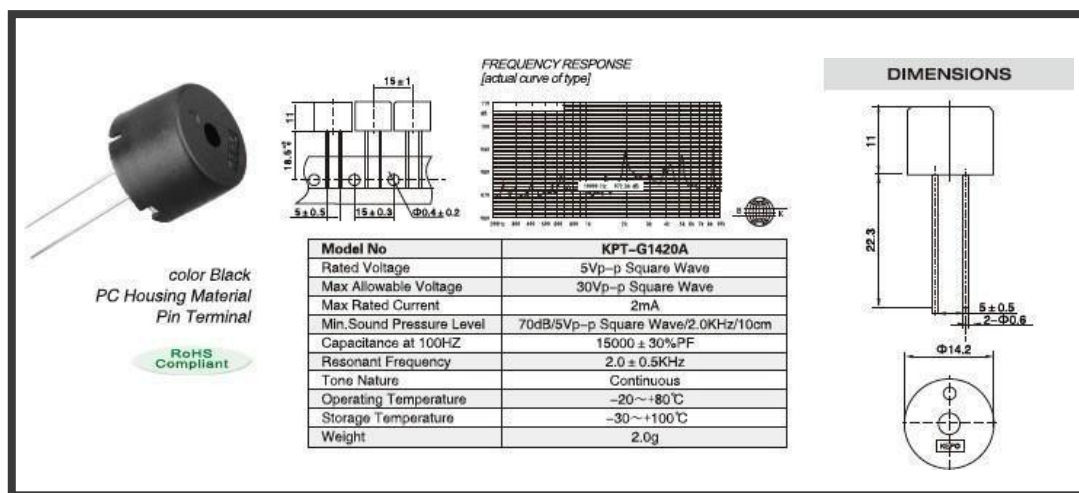


FIG.12 BUZZER

11. Battery

An electric battery is a source of electric power consisting of one or more electrochemical cells with external connections[1] for powering electrical devices. When a battery is supplying power, its positive terminal is the cathode and its negative terminal is the anode.[2] The terminal marked negative is the source of electrons that will flow through an external electric circuit to the positive terminal. When a battery is connected to an external electric load, a redox reaction converts high-energy reactants to lower-energy products, and the free-energy difference is delivered to the external circuit as electrical energy.



FIG.13 BATTERY PACK

12. Pulse Sensor

A pulse sensor works by shining a green light on your finger and measuring the amount of reflected light with a photo sensor. It detects changes in blood volume caused by the heart pumping blood. The redder the blood (higher the hemoglobin level), the greener light it absorbs¹. There are four main ways to measure heart rate: electrocardiogram, photoelectric pulse wave, blood pressure measurement, and phonocardiography.

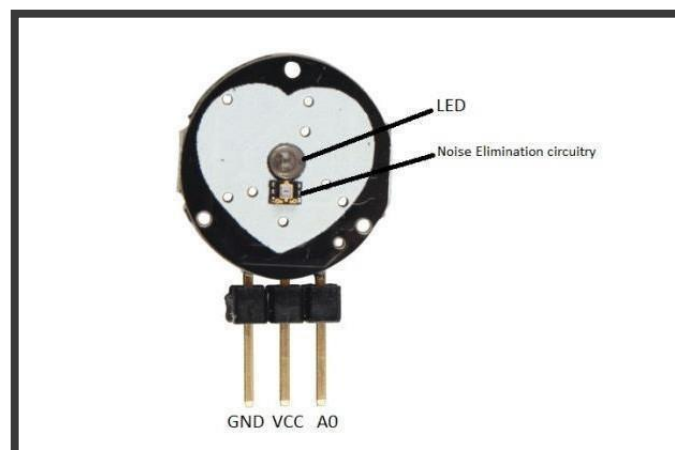


FIG.14 PULSE SENSOR

D. Circuit Diagram

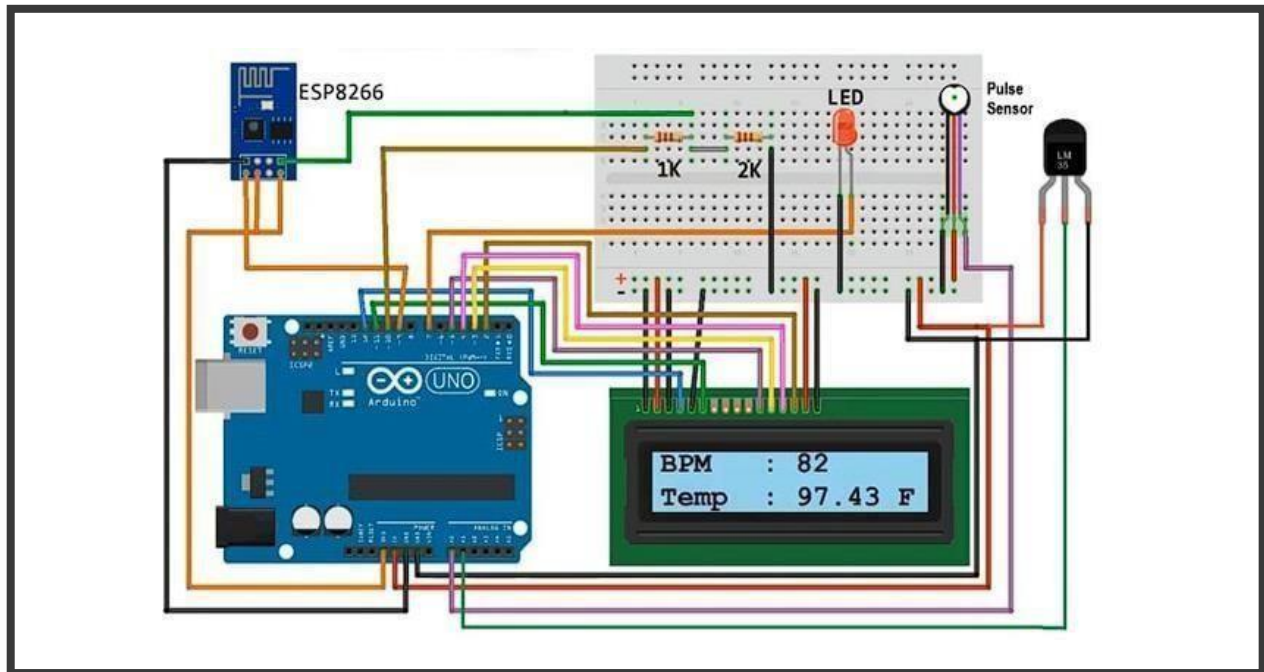


FIG.15 CIRCUIT DIAGRAM

Healthcare Mechanism:

The healthcare mechanism of the elderly care robot utilizes Arduino IoT capabilities to monitor vital signs and provide basic healthcare assistance. Arduino boards, integrated with sensors such as the DHT11 temperature and humidity sensor and the pulse sensor, collect real-time data

on the user's health status. This information is displayed on an LCD display for easy viewing and analysis. The ESP8266 Wi-Fi module enables data transmission to external devices or servers, facilitating remote monitoring by caregivers or healthcare professionals.

Through continuous monitoring of vital signs and timely alerts in case of irregularities, the healthcare mechanism ensures the well-being and safety of elderly users. By leveraging Arduino technology, the healthcare mechanism offers a cost-effective and accessible solution to address the healthcare needs of the elderly population.

E. FLOW CHART

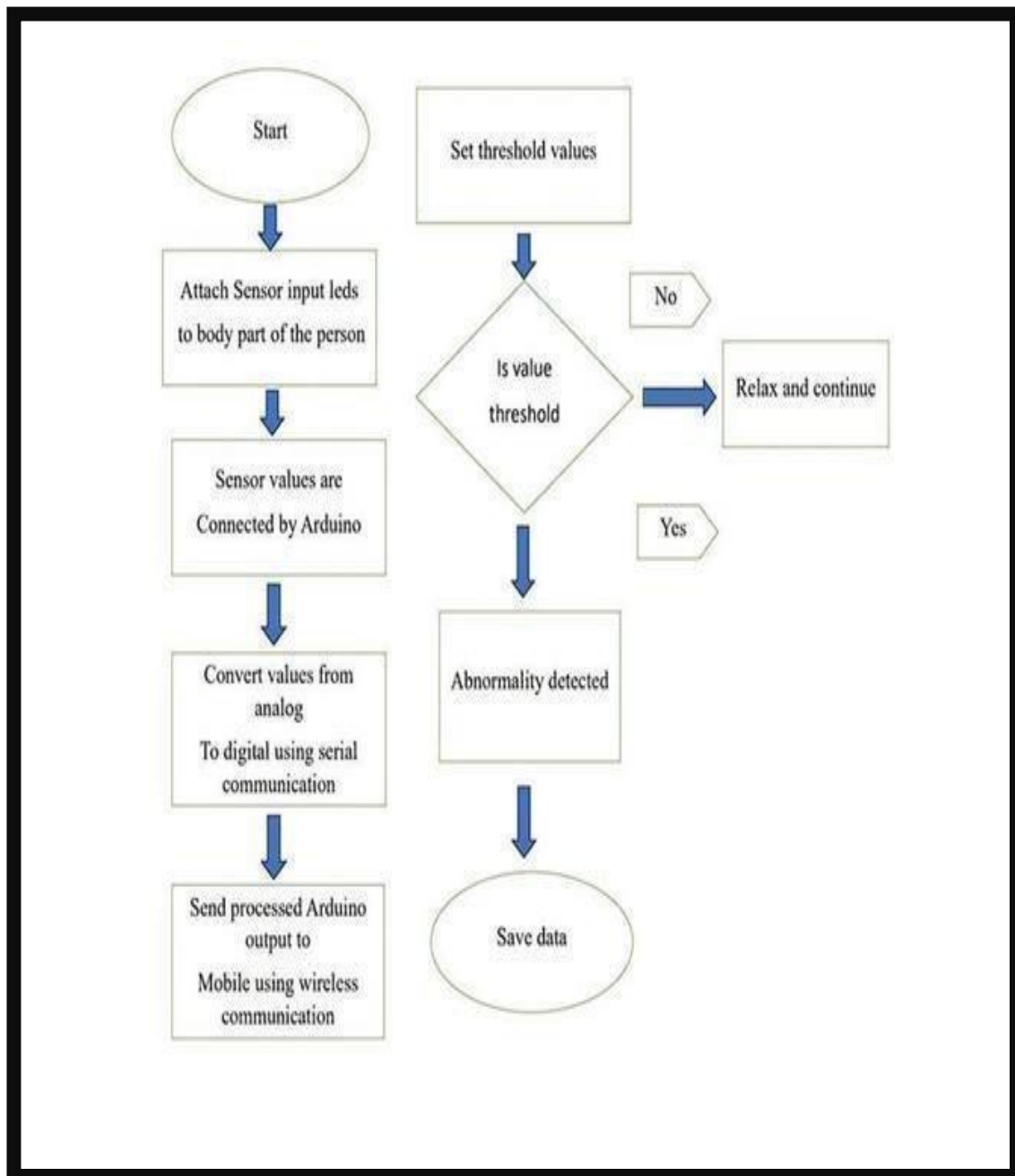


FIG.16 FLOW CHART OF SYSTEM

EXPERIMENTL **VALIIDATION/** **RESULT**

Phase I Result

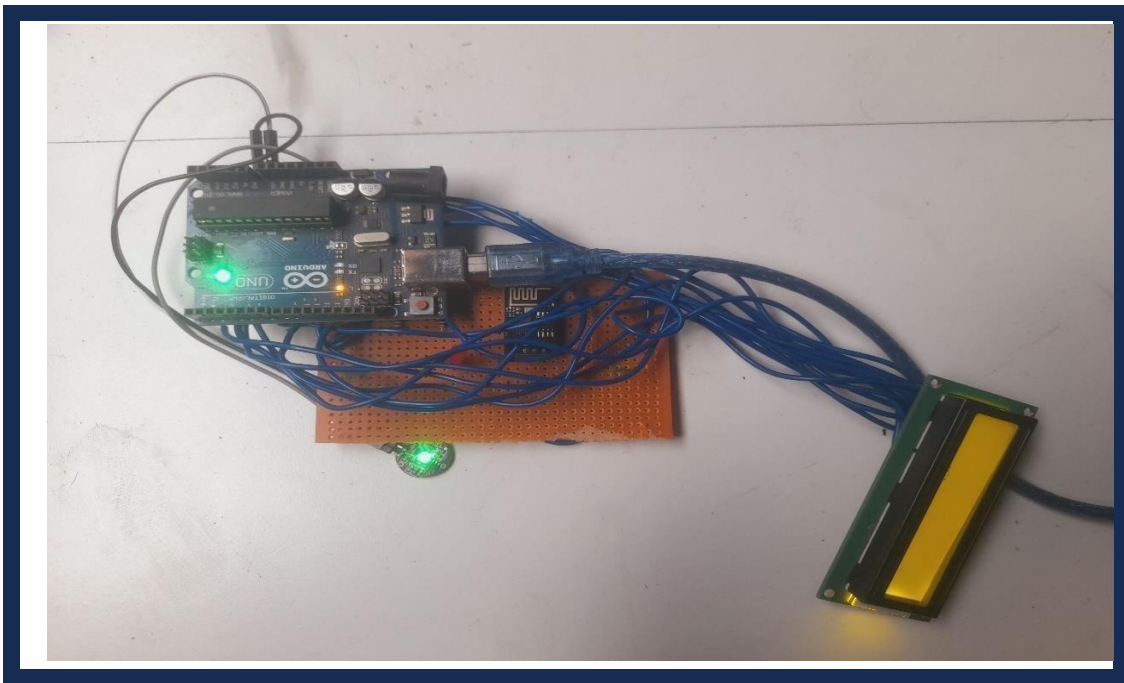
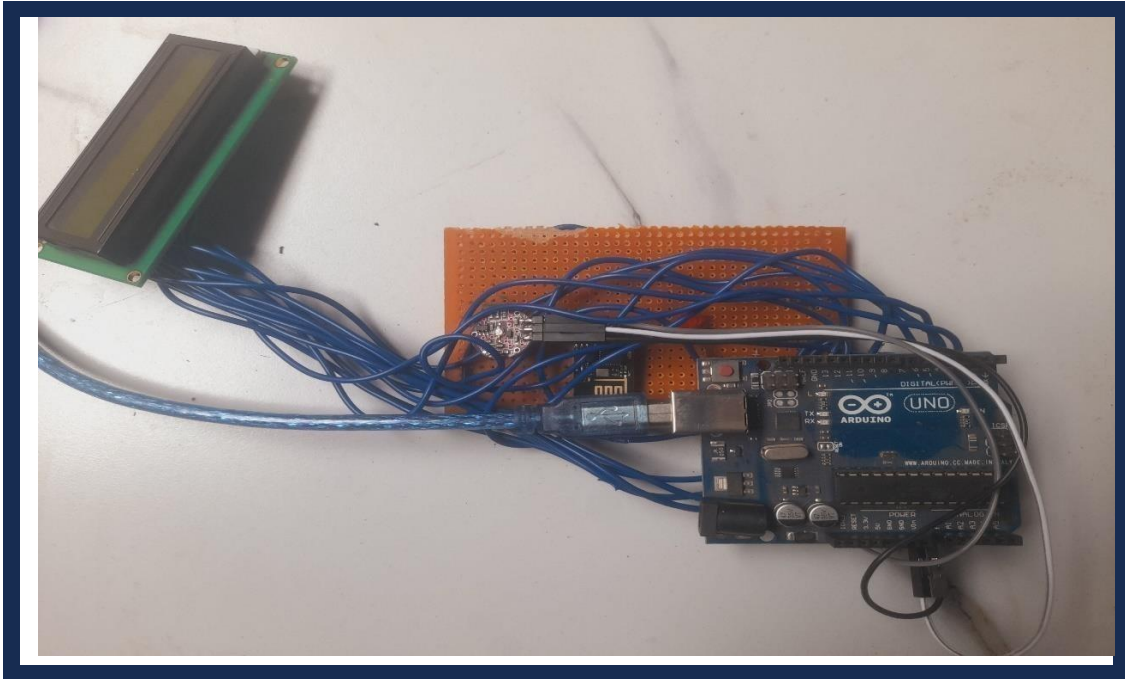


FIG.17 PHASE I AND PHASE II RESULT

Phase II Result



Fig.18 WORKING MODEL

Eldercare robots are intended to help the elderly perform different chores, enhance their quality of life, and give companionship. Here's a summary of how they operate: Sensors and

Perception: Eldercare robots are outfitted with a variety of sensors, including cameras, infrared sensors, and depth sensors, to sense their surroundings and recognize items, barriers, and humans.

Mobility: The majority of eldercare robots are mobile and can roam autonomously or semi-autonomously indoors. They employ mapping and localization techniques to navigate safely and avoid collisions with obstructions.

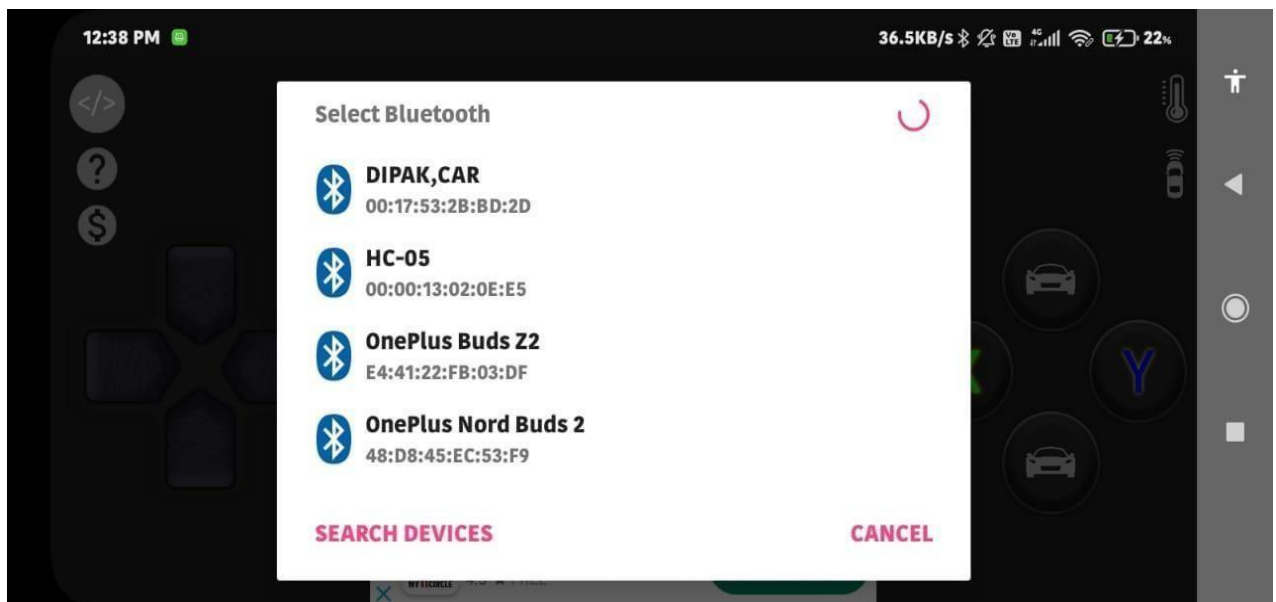
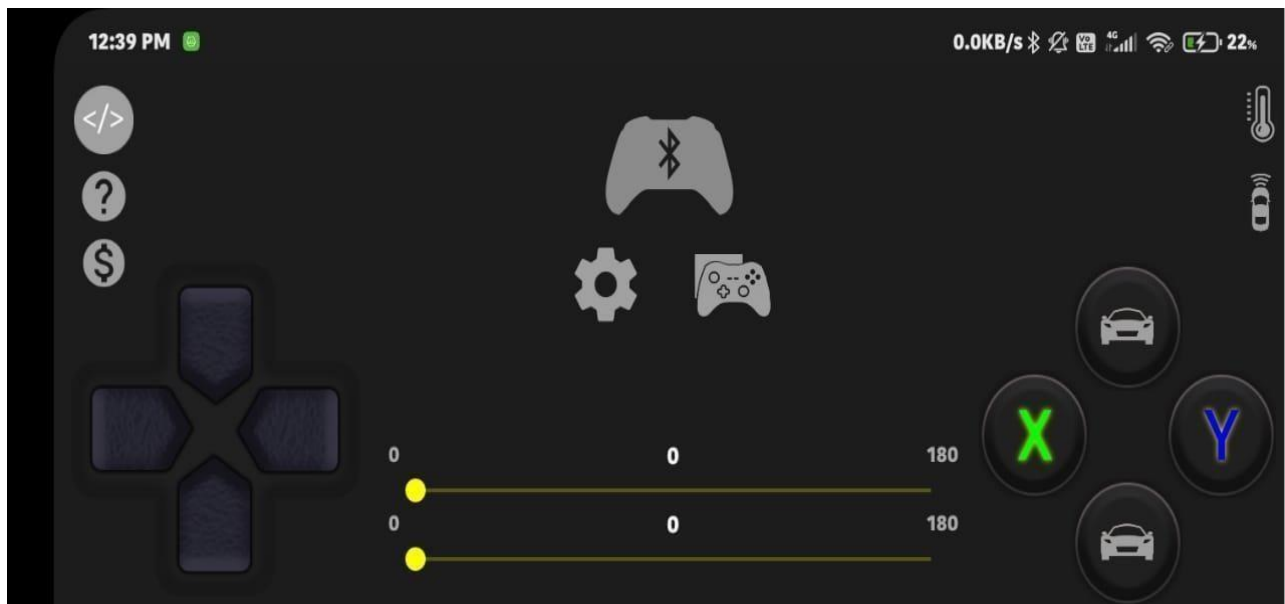
Communication: These robots are frequently outfitted with speakers, microphones, and displays to aid communication with senior people. They can recognize voice commands, answer to queries, and set reminders for medications, appointments, and everyday duties.

Monitoring and Health Tracking: Some eldercare robots are equipped with biometric sensors to monitor vital signs such as heart rate, blood pressure, and body temperature. They can also track activity levels and detect falls. This information can be relayed to caregivers or healthcare professionals for timely intervention if necessary.

Remote Monitoring and Control: Caregivers or family members can often remotely monitor the activities of the eldercare robot through a smartphone app or web interface. They can check on the well-being of their loved ones, receive alerts in case of emergencies, and even remotely control the robot if needed.

Safety Features: Eldercare robots are designed with safety in mind. They have features such as obstacle avoidance to prevent collisions, emergency stop buttons, and fail-safe mechanisms to ensure they operate safely around elderly users.

Overall, eldercare robots aim to enhance the independence, safety, and well-being of the elderly population while providing peace of mind to their caregivers and family members. However it's important to consider ethical and privacy implications when integrating such technology into the care of vulnerable individuals.



SOFTWARE INTERFACE DISPLAY

FIG.19 SNAPSHOT OF RESULT

6. **CONCLUSION**

the development of this elderly care robot represents a significant step forward in addressing the needs of aging populations. Through the integration of Arduino-based hardware and software components, including sensors, displays, and motor controls, the robot offers personalized assistance and healthcare monitoring tailored to individual users. The compact and modular design ensures ease of assembly, maintenance, and customization, while emphasizing safety, mobility, and user interaction. Calibration and testing procedures validate the reliability and accuracy of the robot's functionalities, while optimization efforts strive to enhance performance and efficiency. With its user-friendly interface and IoT-enabled capabilities, the robot stands poised to make a meaningful impact in improving the quality of life for elderly individuals and their caregivers. As technology continues to evolve, this project lays the foundation for future advancements in elder care robotics, promising continued innovation and progress in the field. Companion robots based on emotional interaction have social and companionship capabilities, effectively assisting elderly individuals in addressing psychological and emotional problems such as isolation, emotion, stress, and loneliness, thereby improving their well-being.

7. SCOPE FOR THE FUTURE WORK

1. **Remote Health Monitoring:** The robot can be equipped with sensors to remotely monitor vital signs and medication adherence, providing real-time health updates to healthcare providers and family members.
2. **Emergency Response:** With built-in fall detection and emergency call features, the robot can promptly respond to emergencies by alerting caregivers and providing assistance to elders in distress.
3. **Medication Management:** Utilizing AI algorithms, the robot can remind elders to take medication on time, dispense pills, and maintain a medication schedule, ensuring proper adherence and reducing medication errors.
4. **Companion and Entertainment:** Incorporating conversational AI and entertainment features, the robot can engage elders in conversation, play music, and suggest activities to combat loneliness and improve mental well-being.
5. **Physical Assistance:** Equipped with robotic arms and lifting mechanisms, the robot can assist elders with mobility challenges in daily activities such as getting out of bed, sitting down, and walking, promoting independence and safety.
6. **Health Education:** The robot can provide personalized health education and exercise guidance tailored to the individual needs of elders, promoting healthy habits and disease prevention.
7. **Environmental Monitoring:** Fitted with environmental sensors, the robot can monitor indoor air quality, temperature, and humidity levels, alerting caregivers to potential hazards and ensuring a safe living environment.
8. **Social Connection:** Through video calls and social networking features, the robot can facilitate communication between elders and their family members, friends, and caregivers, fostering social connections and emotional support.
9. **Meal Assistance:** Incorporating meal planning and cooking functionalities, the robot can help elders plan nutritious meals, prepare ingredients, and even assist in cooking, ensuring proper nutrition and mealtime enjoyment.

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SOURCE **CODE**

9. SOURCE CODE

- **Serving Mechanism**

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
const int trigPin = 3; // Trig pin of the sensor connected to digital pin 2
const int echoPin = 2; // Echo pin of the sensor connected to digital pin 3

// Variables to store the duration and distance
int b = 0;
int c = 0;
#define A1 5
#define A2 6
#define B3 7
#define B4 8
#define C1 9
#define C2 10
long duration;
int distance;
void forward(){
    digitalWrite(A1,0);
    digitalWrite(A2,1);
    digitalWrite(B3,1);
    digitalWrite(B4,0);
}
void back(){
    digitalWrite(A1,1);
    digitalWrite(A2,0);
    digitalWrite(B3,0);
    digitalWrite(B4,1);
}
void right(){
    digitalWrite(A1,1);
    digitalWrite(A2,0);
    digitalWrite(B3,1);
    digitalWrite(B4,0);
}
void left(){
    digitalWrite(A1,0);
    digitalWrite(A2,1);
    digitalWrite(B3,0);
    digitalWrite(B4,1);
}
```

```

void motorstop(){
    digitalWrite(A1,1);
    digitalWrite(A2,1);
    digitalWrite(B3,1);

    digitalWrite(B4,1);
}
void upleft(){
    digitalWrite(C1,1);
    digitalWrite(C2,0);
}
void upright(){
    digitalWrite(C1,0);
    digitalWrite(C2,1);
}
void uprightstop(){
    digitalWrite(C1,1);
    digitalWrite(C2,1);
}
void upleftstop(){
    digitalWrite(C1,1);
    digitalWrite(C2,1);
}
#define buzzer 4
char a;
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
// declare an SSD1306 display object connected to I2C
Adafruit_SSD1306 oled(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
void distance1(){
    Serial.println("object ahead");
    oled.clearDisplay();
    oled.setTextSize(2);      // text size
    oled.setTextColor(WHITE); // text color
    oled.setCursor(30, 20);    // position to display
    oled.println("Object");
    oled.setCursor(30, 40);    // position to display
    oled.println("Ahead");
    oled.display();
}
void setup() {

```

```

    pinMode(C1,OUTPUT);
    pinMode(C2,OUTPUT);
    pinMode(A1,OUTPUT);
    pinMode(A2,OUTPUT);
    pinMode(B3,OUTPUT);
    pinMode(B4,OUTPUT);
    // put your setup code here, to run once:
    // initialize OLED display with address 0x3C for 128x64
    if (!oled.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
        Serial.println(F("SSD1306 allocation failed"));
        while (true);
    }
    Serial.begin(9600);
    pinMode(buzzer,OUTPUT);

    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
}
void loop() {
    // put your main code here, to run repeatedly:
    digitalWrite(trigPin, LOW);
    delayMicroseconds(10);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;
    if(distance <= 30){
        digitalWrite(buzzer,HIGH);
        distance1();
    }
    else{
        digitalWrite(buzzer,LOW);
        oled.clearDisplay();
        oled.setTextSize(2);
        oled.setTextColor(SSD1306_WHITE);
        oled.setCursor(0,0);
        oled.print("Distance:");
        oled.setCursor(0,30);
        oled.print(distance);
        oled.print(" cm");
        oled.display();
    }
}

```

```

if(Serial.available()){
  char A = Serial.read();
  Serial.println(A);
  if(A == 'F'){
    Serial.println("Fruunt");
    forward();
  }
  else if( A == 'G'){
    Serial.println("back");
    back();
  }
  else if( A == 'R'){
    Serial.println("right");
    right();
  }
  else if( A == 'L'){
    Serial.println("left");
    left();
  }
  else if( A == 'S'){
    Serial.println("stop");
    motorstop();

  }
  else if( A == 'Y' && b == 0){
    Serial.println("rightUp");
    upright();
    b = 1;
  }
  else if( A == 'Y' && b == 1){
    Serial.println("rightUpstop");
    uprightstop();
    b = 0;
  }
  else if( A == 'X' && c == 0){
    Serial.println("leftUp");
    upleft();
    c = 1;
  }
  else if( A == 'X' && c == 1){
    Serial.println("leftUpstop");
    upleftstop();
    c = 0;
  }
}

```

```
    else if( A == 'N'){
Serial.println("leftUp");
    }
}
}
```

- **Health Care Mechanism**

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
#include <SoftwareSerial.h>
float pulse = 0;
float temp = 0;
SoftwareSerial ser(9, 10);
String apiKey = "OO707TGA1BLUNN12";
```

```
int pulsePin = A0;
int blinkPin = 7;
int fadePin = 13;
int fadeRate = 0;
```

```
volatile int BPM;
volatile int Signal;
volatile int IBI = 600;
volatile boolean Pulse = false;
volatile boolean QS = false;
```

```
volatile int rate[10];
volatile unsigned long sampleCounter = 0;
volatile unsigned long lastBeatTime = 0;
volatile int P = 512;
volatile int T = 512;
volatile int thresh = 525;
volatile int amp = 100;
volatile boolean firstBeat = true;
volatile boolean secondBeat = false;
```

```
void setup() {
    lcd.begin(16, 2);
    pinMode(blinkPin, OUTPUT);
    pinMode(fadePin, OUTPUT);
}
```

```

Serial.begin(115200);
interruptSetup();
lcd.clear();
lcd.setCursor(0, 0);
lcd.print(" Patient Health");
lcd.setCursor(0, 1);
lcd.print(" Monitoring ");
delay(4000);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Initializing. ..");
delay(5000);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Getting Data....");
ser.begin(9600);
ser.println("AT");
delay(1000);
ser.println("AT+GMR");
delay(1000);
ser.println("AT+CWMODE=3");
delay(1000);
ser.println("AT+RST");
delay(5000);
ser.println("AT+CIPMUX=1");
delay(1000);

String cmd = "AT+CWJAP=\"Alexahome\", \"98765432\"";
ser.println(cmd);
delay(1000);
ser.println("AT+CIFSR");
delay(1000);
}

void loop() {

  serialOutput();
  if (QS == true) {
    fadeRate = 255;
    serialOutputWhenBeatHappens();
    QS = false;
  }
}

```

```

    ledFadeToBeat();
    delay(20);
    read_temp();
    esp_8266();
}
void ledFadeToBeat() {
    fadeRate -= 15;
    fadeRate = constrain(fadeRate, 0, 255);
    analogWrite(fadePin, fadeRate);
}

void interruptSetup() {
    TCCR2A = 0x02;
    TCCR2B = 0x06;
    OCR2A = 0X7C;
    TIMSK2 = 0x02;
    sei();
}

void serialOutput() {
    if (serialVisual == true) {
        arduinoSerialMonitorVisual('-', Signal);
    } else {
        sendDataToSerial('S', Signal);
    }
}

void serialOutputWhenBeatHappens() {
    if (serialVisual == true) {
        Serial.print("**** Heart-Beat Happened **** ");
        Serial.print("BPM: ");
        Serial.println(BPM);
    } else {
        sendDataToSerial('B', BPM);
        sendDataToSerial('Q', IBI);
    }
}

void arduinoSerialMonitorVisual(char symbol, int data) {
    const int sensorMin = 0;
    const int sensorMax = 1024;
    int sensorReading = data;
    int range = map(sensorReading, sensorMin, sensorMax, 0, 11);

    switch (range) {

```

```

case 0:
    Serial.println("");
    break;
case 1:
    Serial.println("---");
    break;
case 2:
    Serial.println("---- ");
    break;
case 3:
    Serial.println("----- ");
    break;
case 4:
    Serial.println("----- ");
    break;
case 5:
    Serial.println("-----|-");
    break;
case 6:
    Serial.println("-----| -- ");
    break;
case 7:
    Serial.println("-----|----- ");
    break;
case 8:
    Serial.println(" .....| .....");
    break;
case 9:
    Serial.println(" .....| .....");
    break;
case 10:
    Serial.println(" .....| .....");
    break;
case 11:
    Serial.println(" .....| .....");
    break;
}
}

void sendDataToSerial(char symbol, int data) {
    Serial.print(symbol);
    Serial.println(data);
}

```

```

ISR(TIMER2_COMPA_vect) {
    cli();
    Signal = analogRead(pulsePin);
    sampleCounter += 2;
    int N = sampleCounter - lastBeatTime;

    if (Signal < thresh && N > (IBI/5)*3) {
        if (Signal < T) {
            T = Signal;
        }
    }
    if (Signal > thresh && Signal > P) {
        P = Signal;
    }
    if (N > 250) {
        if ( (Signal > thresh) && (Pulse == false) && (N > (IBI/5)*3) ) {
            Pulse = true;
            digitalWrite(blinkPin, HIGH);
            IBI = sampleCounter - lastBeatTime;
            lastBeatTime = sampleCounter;

            if(secondBeat) {
                secondBeat = false;
                for(int i=0; i<=9; i++) {
                    rate[i] = IBI;
                }
            }
            if(firstBeat) {
                firstBeat = false;
                secondBeat = true;
                sei();
                return;
            }
            word runningTotal = 0;
            for(int i=0; i<=8; i++) {
                rate[i] = rate[i+1];
                runningTotal += rate[i];
            }
            rate[9] = IBI;
            runningTotal += rate[9];
            runningTotal /= 10;
            BPM = 60000/runningTotal;
            QS = true;
            pulse = BPM;
        } }
    }

```

```

if (Signal < thresh && Pulse == true) {
    digitalWrite(blinkPin, LOW);
    Pulse = false;
    amp = P - T;
    thresh = amp/2 + T;
    P = thresh;
    T = thresh;
}
if (N > 2500) {
    thresh = 512;
    P = 512;
    T = 512;

    lastBeatTime = sampleCounter;
    firstBeat = true;
    secondBeat = false;
}
sei();
}
void esp_8266() {
    String cmd = "AT+CIPSTART=4,\"TCP\", \"\"";
    cmd += "184.106.153.149";
    cmd += "\",80";
    ser.println(cmd);
    Serial.println(cmd);
    if(ser.find("Error")) {
        Serial.println("AT+CIPSTART error");
        return;
    }
    String getStr = "GET /update?api_key=";
    getStr += apiKey;
    getStr += "

```

10. **RESULT**

1. The robot provides real-time monitoring of vital signs, such as pulse and temperature, ensuring accurate and timely healthcare data collection
2. A user-friendly interface allows users and caregivers to easily access and interpret healthcare information and interact with the robot,
3. The robot successfully integrates IoT capabilities for remote monitoring and control, enhancing user safety and convenience
4. The robot effectively serves meals, drinks, and assists with daily tasks, promoting independence and comfort for elderly users
5. The robot incorporates obstacle detection mechanisms, ensuring safe navigation and reducing the risk of collisions
6. It features emergency shutdown options, adding an extra layer of safety and enabling immediate intervention when necessary
7. The robot complies with data privacy regulations and safeguards sensitive healthcare data to protect user privacy
8. An effective alert system is in place to notify caregivers and users of health anomalies or emergencies.
9. The robot's performance is continuously refined based on user feedback and iterative development, enhancing its overall effectiveness and user satisfaction.

APPENDIX

11. APPENDIX

Sr No	Item	Amount
1.	Arduino Uno	765 Rs
2.	Arduino Nano 3,0	550 Rs
3.	LCD Display	203 Rs
4.	L293D motor driver (2)	335 Rs
5.	OLED Display	349 Rs
6.	ESP8266 Wi-fi	216 Rs
7.	HC-05	650 Rs
8.	DC motors (5)	1500 Rs
9.	Ultrasonic Sensor	350 Rs
10.	DHT-11 temperature sensor	150 Rs
11.	Pulse sensor	250 Rs
12.	Battery	1560 Rs
13.	Wheels	660 Rs
Total		7538 Rs

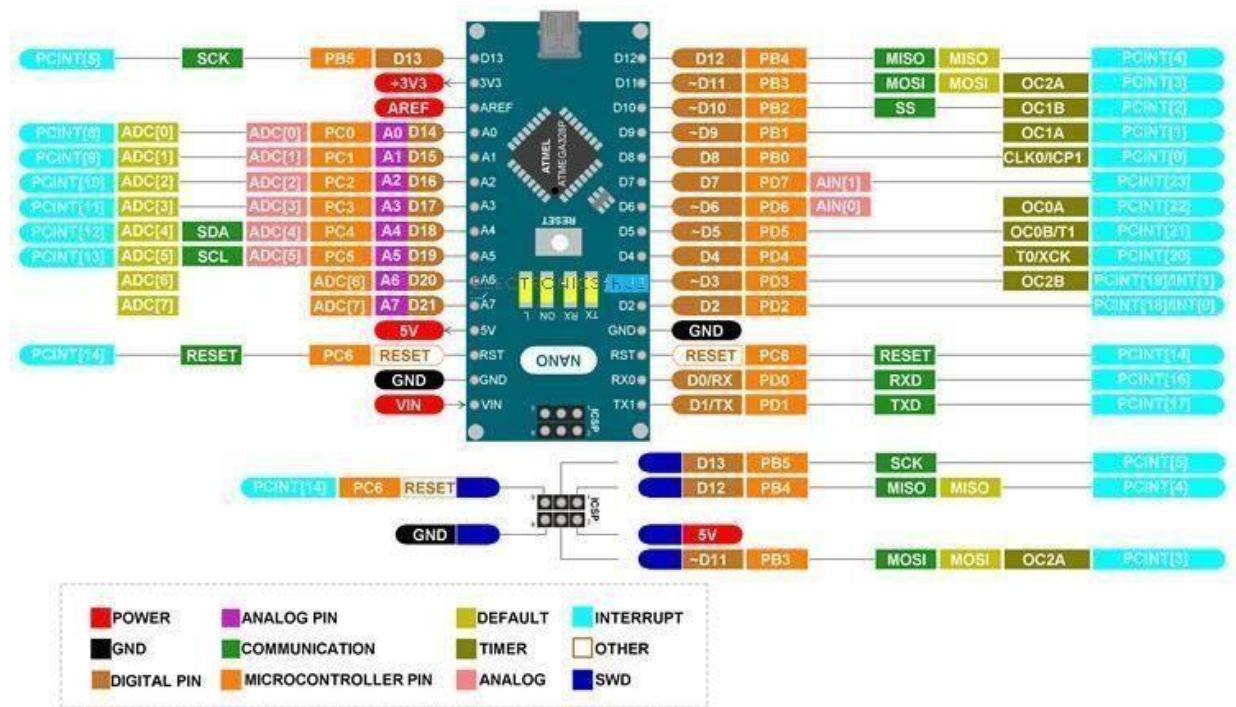
1. Arduino Nano :-

Description :

The Arduino Nano is an open-source breadboard-friendly microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2008. It offers the same connectivity and specs of the Arduino Uno board in a smaller form factor. The Arduino Nano is equipped with 30 male I/O headers, in a DIP-30-like configuration, which can be programmed using the Arduino Software integrated development environment (IDE), which is common to all Arduino boards and running both online and offline. The board can be powered through a type-B mini-USB cable or from a 9V battery

Rather than requiring a physical press of the reset button before an upload, the Arduino Nano is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the FT232RL is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor.

When this line is asserted (taken low), the reset line drops long enough to reset the chip. This setup has other implications. When the Nano is connected to a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the Nano. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened.



Powering Arduino Nano:

There are total three ways by which you can power your Nano.

USB Jack: Connect the mini USB jack to a phone charger or computer through a cable and it will draw power required for the board to function

Vin Pin: The Vin pin can be supplied with an unregulated 6-12V to power the board. The on-board voltage regulator regulates it to +5V.

+5V Pin: If you have a regulated +5V supply then you can directly provide this to the +5V pin of the Arduino.

Input/output:

There are total 14 digital Pins and 8 Analog pins on your Nano board. The digital pins can be used to interface sensors by using them as input pins or drive loads by using them as output pins. A simple function like **pinMode()** and **digitalWrite()** can be used to control their operation. The operating voltage is 0V and 5V for digital pins. The analog pins can measure analog voltage from 0V to 5V using any of the 8 Analog pins using a simple function like **analogRead()**.

These pins apart from serving their purpose, can also be used for special purposes, which are discussed below:

Serial Pins 0 (Rx) and 1 (Tx): Rx and Tx pins are used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.

External Interrupt Pins 2 and 3: These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.

PWM Pins 3, 5, 6, 9 and 11: These pins provide an 8-bit PWM output by using **analogWrite()** function.

SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK): These pins are used for SPI communication.

In-built LED Pin 13: This pin is connected with a built-in LED. When pin 13 is HIGH – LED is on and when pin 13 is LOW, it is off.

I2C A4 (SDA) and A5 (SCA): Used for IIC communication using Wire library.

AREF: Used to provide reference voltage for analog inputs with **analogReference()** function.

Reset Pin: Making this pin LOW, resets the microcontroller

Digital Pins :

Digital pins can be used as both input and output pins on most Arduino microcontrollers. This means that a single pin can both send and receive digital data. Digital Arduino pins can be only "high" or "low." When a digital pin is high, a constant voltage is present. When the pin is low, this voltage is gone. Switching a light emitting diode (LED) on and off is one common instance of a digital output, and checking the status of a button is an example of a digital input.

Analog Pins :

Analog input pins are used to read the state of a connected device, such as a sensor. Although digital pins can only tell if a connection is high or low, analog input pins are able to read the exact voltage value. This is useful for interfacing with analog sensors that have a wide range of

possible values. For example, analog thermometer sensors often send a voltage reading that corresponds to the temperature

Power Pins :

Power Pins Most Arduino microcontrollers have several power pins that receive and output a variety of voltages. The Vin, 5V, 3.3V, and GND pins are Arduino power pins. You can use the Vin pin to Power your Arduino with an unregulated 7 to 12-volt power source. Like a 9V battery or a wall adapter that is in the range of 7 to 12 volts.

2. Ultrasonic Sensor :

Ultrasonic sensors use sound waves to detect distance and motion, and can have a range of several meters. They are typically small and lightweight, made of high quality materials, and have a long service life. Some features of ultrasonic sensors include:

- **Range:** Can detect distances from 20 mm to 8 m
- **Accuracy:** Can penetrate deep into materials or components
- **Angle:** Can emit ultrasonic waves that reflect off the device, and the strength and angle of the waves reflected can determine the object's position
- **Chemical resistant:** Can be used in all weather conditions and are resistant to chemicals in pesticides
- **Operating modes:** Typically operate in distance-to-object (DtO) mode, which sends a signal when an object is within range

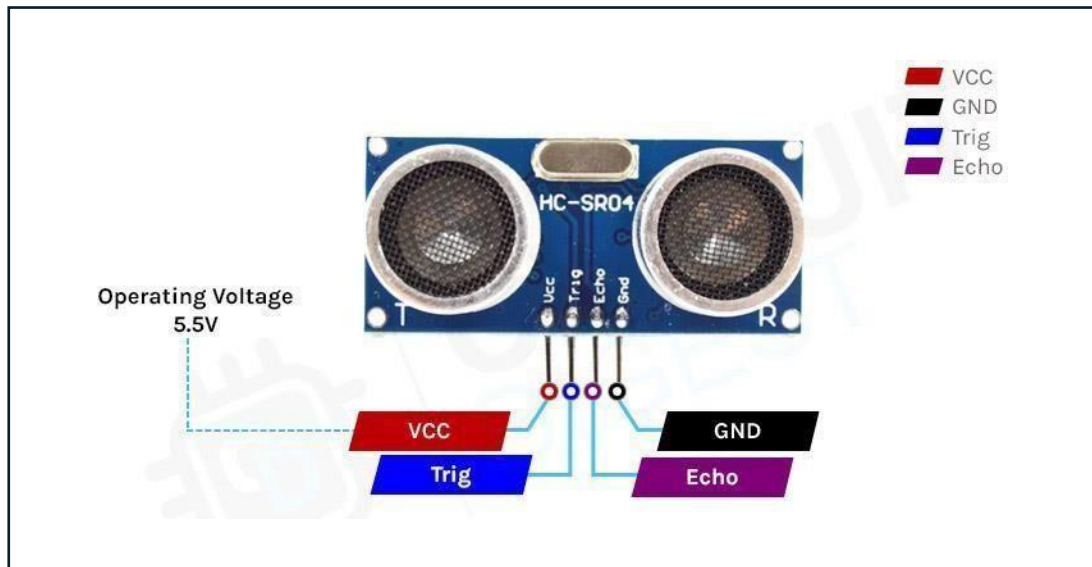


FIG.21 PIN OUT FOR ULTRASONIC SENSOR

Pin Number	Pin Name	Description
1	Vcc	The Vcc pin powers the sensor, typically with +5V
2	Trigger	Trigger pin is an Input pin. This pin has to be kept high for 10us to initialize measurement by sending US wave.
3	Echo	Echo pin is an Output pin. This pin goes high for a period of time which will be equal to the time taken for the US wave to return back to the sensor.
4	Ground	This pin is connected to the Ground of the system.

TABLE.7 PIN OUT FOR ULTRASONIC SENSOR

3. **HC-05 (Bluetooth Module) :**

It is used for many applications like wireless headset, game controllers, wireless mouse, wireless keyboard, and many more consumer applications.

It has range up to <100m which depends upon transmitter and receiver, atmosphere, geographic & urban conditions.

It is IEEE 802.15.1 standardized protocol, through which one can build wireless Personal Area Network (PAN). It uses frequency-hopping spread spectrum (FHSS) radio technology to send data over air.

It uses serial communication to communicate with devices. It communicates with microcontroller using serial port (USART).

Bluetooth serial modules allow all serial enabled devices to communicate with each other using Bluetooth.

It has 6 pins,

Key/EN: It is used to bring Bluetooth module in AT commands mode. If Key/EN pin is set to high, then this module will work in command mode. Otherwise by default it is in data mode. The default baud rate of HC-05 in command mode is 38400bps and 9600 in data mode.

VCC: Connect 5 V or 3.3 V to this Pin.

GND: Ground Pin of module.

TXD: Transmit Serial data (wirelessly received data by Bluetooth module transmitted out serially on TXD pin)

RXD: Receive data serially (received data will be transmitted wirelessly by Bluetooth module).

State: It tells whether module is connected or not

HC-05 module has two modes

Data mode: Exchange of data between devices.

Command mode: It uses AT commands which are used to change setting of HC-05. To send these commands to module serial (USART) port is used.

Features Of HC05 :

Bluetooth version: 2.0 + EDR (Enhanced Data Rate)

Frequency: 2.4 GHz ISM band

Modulation: GFSK (Gaussian Frequency Shift Keying)

Transmit power: Class 2 (up to 4 dBm)

Sensitivity: -80 dBm typical

Range: approximately 10 meters (or 33 feet) in open air

Profiles supported: SPP (Serial Port Profile), HID (Human Interface Device) and others

Operating voltage: 3.3V to 5V DC

Operating current: less than 50mA

4. ESP8266 (Wifi Module) :

An ESP8266 Wi-Fi module is a SOC microchip mainly used for the development of end-point IoT (Internet of things) applications. It is referred to as a standalone wireless transceiver, available at a very low price. It is used to enable the internet connection to various applications of embedded systems. It is a SOC (System On-chip) integrated with a TCP/IP protocol stack, which can provide microcontroller access to any type of Wi-Fi network. This article deals with the pin configuration, specifications, circuit diagram, applications, and alternatives of the ESP8266 Wi-Fi module.

The processor of this module is based on the Tensilica Xtensa Diamond Standard 106 micro and operates easily at 80 MHz. There are different types of ESP modules designed by third-party manufacturers. They are,

ESP8266-01 designed with 8 pins (GPIO pins -2)

ESP8266-02 designed with 8 pins (GPIO pins -3)

ESP8266-03 designed with 14 pins (GPIO pins- 7)

ESP8266-04 designed with 14 pins (GPIO pins- 7)

The ESP8266 Wi-Fi module comes with a boot ROM of 64 KB, user data RAM of 80 KB, and instruction RAM of 32 KB. It can support 802.11 b/g/n Wi-Fi network at 2.4 GHz along with the features of I2C, SPI, I2C interfacing with DMA, and 10-bit ADC. Interfacing this module with the microcontroller can be done easily through a serial port. An external voltage converter is required only if the operating voltage exceeds 3.6 Volts. It is most widely used in robotics and IoT applications due to its low cost and compact size.

Pin Configuration/Pin Diagram

The **ESP8266 Wi-Fi module pin configuration/pin diagram** is shown in the figure below. The ESP8266-01 Wi-Fi module runs in two modes. They are;

Flash Mode: When GPIO-0 and GPIO-1 pins are active high, then the module runs the program, which is uploaded into it.

UART Mode: When the GPIO-0 is active low and GPIO-1 is active high, then the module works in programming mode with the help of either serial communication or Arduino board.

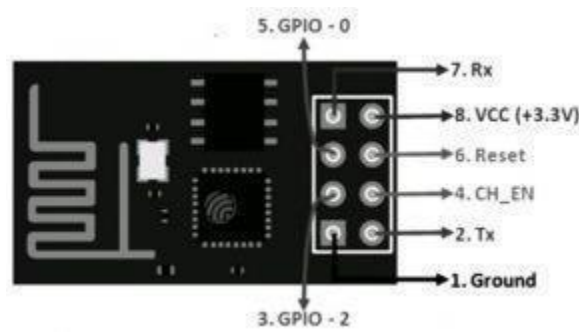


FIG.22 PINOUT FOR ESP8266

5. L293D Motor Driver :

L293D is a basic motor driver integrated chip (IC) that enables us to drive a DC motor in either direction and also control the speed of the motor. The L293D is a 16 pin IC, with 8 pins on each side, allowing us to control the motor. It means that we can use a single L293D to run up to two DC motors. L293D consist of two H-bridge circuit. H-bridge is the simplest circuit for changing polarity across the load connected to it. There are 2 OUTPUT pins, 2 INPUT pins, and 1 ENABLE pin for driving each motor. It is designed to drive inductive loads such as solenoids, relays, DC motors, and bipolar stepper motors, as well as other high-current/high-voltage loads.

Technical Specification:

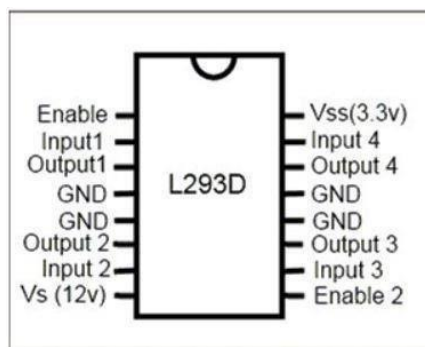
- Wide Supply-Voltage Range: 4.5 V to 36 V
- Output Current: 600 mA Per Channel
- Peak Output Current: 1.2 A Per Channel
- Supply Voltage to the IC: 4.5V to 7V
- Transition time: 300ns (at 5V and 24V)
- Size: 44mm* 40mm*20mm (L*W*H)

Features:

- LM393 based design
- Easy to use with Microcontrollers or even with normal Digital/Analog IC
- Small, cheap and easily available
- Fixed bolt hole for easy installation
- Comparator output, clean signal, great waveform, and strong driving ability

Applications:

- Robotics Applications
- Mechatronics Application



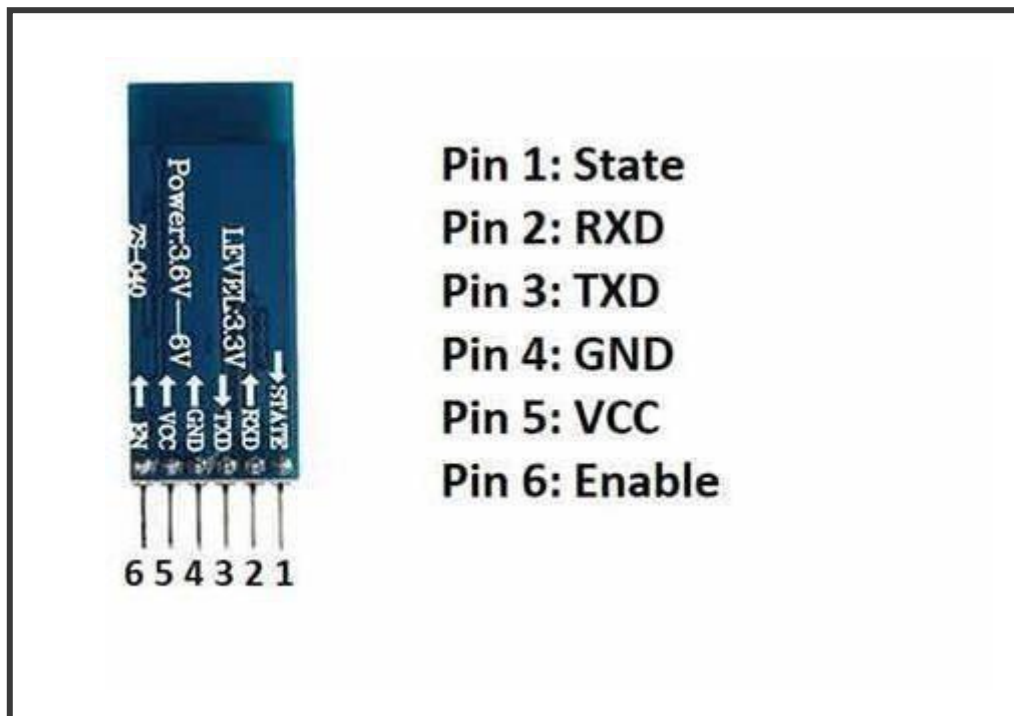
**FIG.24 PINOUT FOR L293D
MOTORDRIVER**

6. HC-05 (Bluetooth Module) :

The HC-05 is a class 2 Bluetooth module designed for transparent wireless serial communication. It is pre-configured as a slave Bluetooth device. Once it is paired to a master Bluetooth device such as PC, smart phones and tablet, its operation becomes transparent to the user. All data received through the serial input is immediately transmitted over the air. When the module receives wireless data, it is sent out through the serial interface exactly at it is received. No user code specific to the Bluetooth module is needed at all in the user microcontroller program. The HC-05 supports two work modes: Command and Data mode. The work mode of the HC-05 can be switched by the onboard push button. The HC-05 is put in Command mode if the push button is activated. In Command mode, user can change the system parameters (e.g. pin code, baud rate, etc) using host controller itself of a PC running terminal software using a serial to TTL converter. Any changes made to system parameters will be retained even after power is removed. Power cycle the HC-05 will set it back to Data Mode. Transparent UART data transfer with a connected remote device occurs only while in Data Mode. The HC-05 can be re-configured by the user to work as a master Bluetooth device using a set of AT commands. Once configured as master, it can automatically pair with a HC-05 in its default slave configuration or a HC-06 module, allowing an point to point serial communications. The HC-05 will work with supply voltage of 3.6VDC to 6VDC, however, the logic level of RXD pin is 3.3V and is not 5V tolerant. A Logic Level Converter is recommended to protect the sensor if connect it to a 5V device (e.g Arduino Uno and Mega). The power to the HC-05 will cut off if the "EN" pin is pulled to logic 0.

Features:

- Bluetooth v2.0+EDR
- 2.4GHz ISM band frequency
- Supported baud rate: 9600 (default), 19200, 38400, 57600, 115200, 230400, 460800.
- Speed: Asynchronous: 2.1Mbps(Max) / 160 kbps, Synchronous: 1Mbps/1Mbps
- Power supply: 3.6V to 6V DC



**FIG.24 PINOUT FOR HC-05
BLUETOOTH MODULE**

RESEARCH
PAPER
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ELDER'S CARE ROBOT

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Abstract: This problem will be solved by the proposed effort, which will create a sophisticated robotic system to help elderly people with daily tasks. As the world's population has grown, there is a greater need for efficient and caring aged care. However, the limited healthcare resources frequently result in insufficient attention and support for elders who need help with daily activities. Additionally, seniors' emotional and social needs are frequently disregarded, which contributes to their feelings of loneliness and despair. Our research aims to develop a companion that can deliver essential services, monitor health, and provide companionship by combining cutting-edge robotics, AI, and human-centered design. The goals, approaches, and anticipated outcomes of our Elder Care Robot project are described in this overview

I. INTRODUCTION

The goal of this project is to create an advanced elder care robot with cutting-edge medical skills and a highly developed serving system. Our mission is to offer comprehensive assistance to the elderly, taking care of their healthcare requirements while boosting their quality of life. Our goal is to enhance the quality of life for our aging population by providing individualized assistance and healthcare services in a practical and effective way.

The proposed work aims to address this challenge by developing an advanced robotic system to assist senior citizens in their daily lives. As the global population has led to an increased demand for effective and compassionate care for the elderly. However, the available healthcare resources are often stretched thin, resulting in inadequate attention and support for seniors who require assistance in their daily lives. Additionally, the emotional and social needs of the elderly are often neglected, leading to feelings of isolation and depression.

Through a combination of state-of-the-art robotics, AI, and human-centered design, our project seeks to create a companion that can provide essential services, monitor health, and offer companionship. This synopsis outlines the objectives, methodologies, and expected outcomes of our Elder Care Robot project, which has the potential to revolutionize elderly care and reshape the way we approach aging societies. Arduino is a microcontroller platform used in healthcare systems for data processing and control. It can collect data from sensors, analyze it, and send commands to other components. Arduino boards are known for their versatility and ease of programming, making them ideal for integrating various sensors and devices. A pulse sensor is a non-invasive device that measures a person's heart rate in real-time. It typically uses an infrared sensor to detect changes in blood volume under the skin, providing accurate pulse rate readings. This data is vital in healthcare applications for monitoring a patient's cardiovascular health and stress levels.

A temperature sensor, such as a digital thermometer or a thermistor, is used to measure a patient's body temperature. This data is crucial for diagnosing and monitoring various medical conditions, especially when detecting fever or hypothermia. Accurate temperature monitoring is essential in healthcare for early intervention. The ESP8266 is a low-cost Wi-Fi module that allows devices to connect to the internet and communicate with other devices or servers.

In a healthcare system, the ESP8266 can transmit sensor data, like pulse and temperature readings, to a central server or display it on a user interface, enabling remote monitoring and analysis, which is particularly valuable for telemedicine and remote patient care. Healthcare is a system that takes care of people's health. It checks our health mainly our pulse and body temperature. The main goal is to keep us healthy and treat elders. It also tries to make sure we can see a doctor when we need to. It's about making sure healthcare is good, safe, and keeps our information private. The big idea is to help everyone stay healthy and get the right care when they need it. TT motors are a type of direct current (DC) motor known for their compact size and high efficiency. They are commonly used in robotic applications due to their ability to provide precise motion control.

In the serving mechanism, TT motors can be employed to control the movement of the serving platform or tray. These motors are usually equipped with rotating shafts that can be coupled with mechanical components to facilitate precise movement. Arduino is a popular microcontroller platform that serves as the brain of the serving mechanism. It provides the necessary processing power to control the TT motors and manage the entire mechanism.

Through Arduino, you can write and upload code to specify how the motors should move, when to start and stop, and how to respond to user commands. The HC-05 module is a Bluetooth communication module that enables wireless control of the serving mechanism. This module can establish a connection with a smartphone or another Bluetooth-enabled device, allowing users to send commands remotely. For instance, you can use a smartphone app to wirelessly control the movement and serving actions of the mechanism. A motor driver is an essential component in the assembly, acting as an intermediary between the Arduino and the TT motors. It helps control the motors' speed and direction, preventing any damage to the motors due to excessive current. A motor driver interprets the signals from the Arduino and amplifies them to drive the motors effectively. It also offers features like motor protection, enabling the safe and reliable operation of the serving mechanism. When these components are integrated into a serving mechanism, we create sophisticated system that allows for precise, remote-controll movement. The Arduino processes user input from the HC-05 Bluetooth module and uses the motor driver to control the TT motors, resulting in the desired serving actions, such as delivering items or moving the serving platform. The assembly combines hardware and software to make the serving mechanism both user-friendly and efficient

II. METHODOLOGY

Developing the "Elder Care Robot with Basic Healthcare and Serving Mechanism" involves defining project objectives, gathering requirements, selecting components, designing and integrating hardware and software, implementing IoT capabilities, testing and ensuring safety, and finally deploying the robot in healthcare settings. Regular user feedback and ongoing improvement efforts are essential aspects of this development process to create an effective and reliable elder care solution.

Hardware Assembly:

I. Prepare Component : Gather all necessary hardware components, including Arduino boards, sensors, displays, motor drivers, and peripherals.

II. Select Chassis : Choose or design a suitable chassis for the robot, ensuring it provides adequate space and support for mounting components.

III. Mount Arduino Boards : Secure the Arduino boards (Uno or Nano) onto the chassis using screws or adhesive, positioning them strategically for easy access and connectivity.

IV. Attach Motor Driver : Mount the L293D motor driver onto the chassis, connecting it to the Arduino board and ensuring proper alignment with the DC motors.

V. Connect DC Motors : Attach the four DC motors to the chassis, positioning them according to the planned serving mechanism configuration and connecting them to the motor driver.

VI. Install Sensors and Displays : Mount the ultrasonic sensor for obstacle detection and OLED display for user feedback onto the chassis, ensuring they are securely attached and positioned for optimal functionality.

VII. Wire Connections : Connect the components using jumper wires or soldering, ensuring correct wiring and polarity to facilitate communication and power distribution.

VIII. Integrate Power Source : Install and connect the battery to power the Arduino boards, motors, and other components, ensuring proper voltage and current ratings for reliable operation.

IX. Test Hardware Setup: Perform initial testing of the hardware assembly to verify component connections, power supply, and mechanical functionality, addressing any issues or discrepancies as needed.

X. Finalize Assembly: Secure all components and connections in place, double-checking for stability and functionality before proceeding to software development and testing.

Software Development:

XI. Setup Arduino IDE: Install and configure the Arduino Integrated Development Environment (IDE) on your computer, ensuring compatibility with the selected Arduino boards and peripherals

XII. Include Libraries : Import necessary libraries for sensor and display modules into the Arduino IDE, enabling access to pre-written functions and code examples for rapid development.

XIII. Define Pin Assignments : Assign digital and analog pins on the Arduino boards for interfacing with sensors, displays, motor driver inputs, and other peripherals, ensuring compatibility and avoiding conflicts.

XIV. Write Initialization Code : Develop initialization code to configure sensor modules, display settings, and motor driver parameters, ensuring proper initialization and calibration before operation.

XV. Implement Sensor Readings : Write code to read sensor data from the DHT11 temperature and humidity sensor and the pulse sensor, converting raw sensor readings into meaningful values for display and analysis. **XVI. Control Motor Movements :** Develop motor control algorithms to drive the DC motors based on input from the ultrasonic sensor and user commands, implementing logic for obstacle detection, navigation, and serving tasks.

XVII. Create User Interface : Design a user-friendly interface using the OLED display to provide feedback and interaction options for users, including menus, prompts, and status indicators for system monitoring.

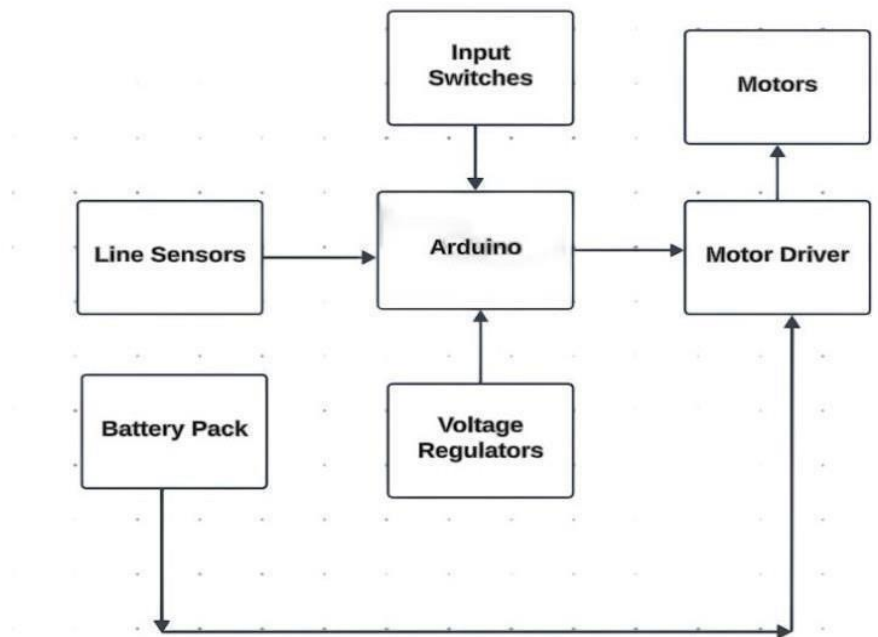
XVIII. Develop Error Handling : Write code to handle errors and exceptions, such as sensor malfunction or communication failures, implementing robust error detection and recovery mechanisms to ensure system reliability.

XIX. Test and Debug : Conduct comprehensive testing of the software code, including unit testing of individual components and integration testing of the entire system, identifying and debugging any errors or inconsistencies encountered during operation.

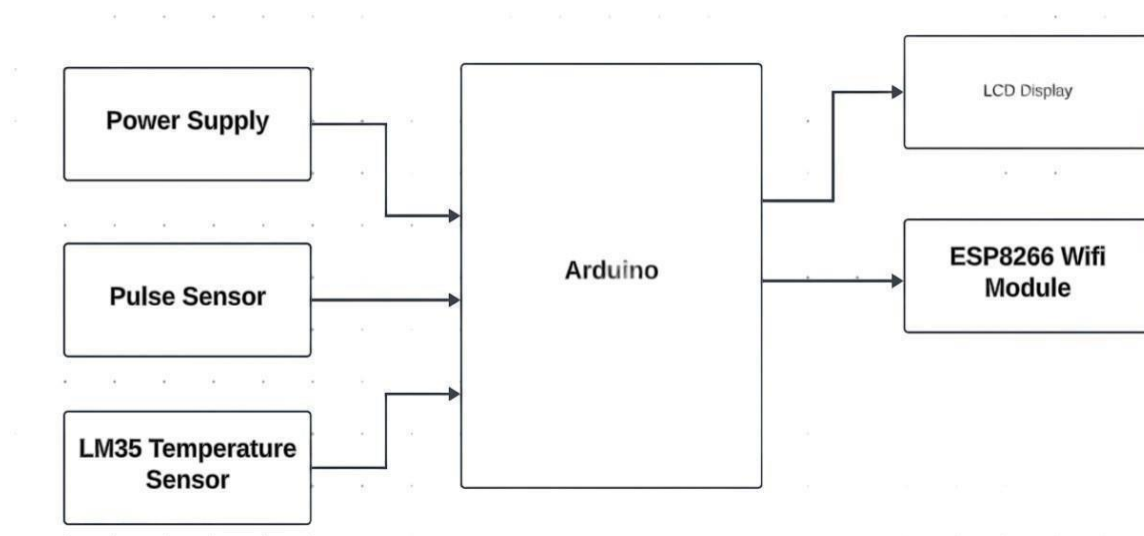
XX. Optimize Code Efficiency : Refactor and optimize the code for efficiency and performance, reducing memory usage, minimizing execution time, and enhancing responsiveness to user inputs and environmental changes.

III. BLOCK DIAGRAM

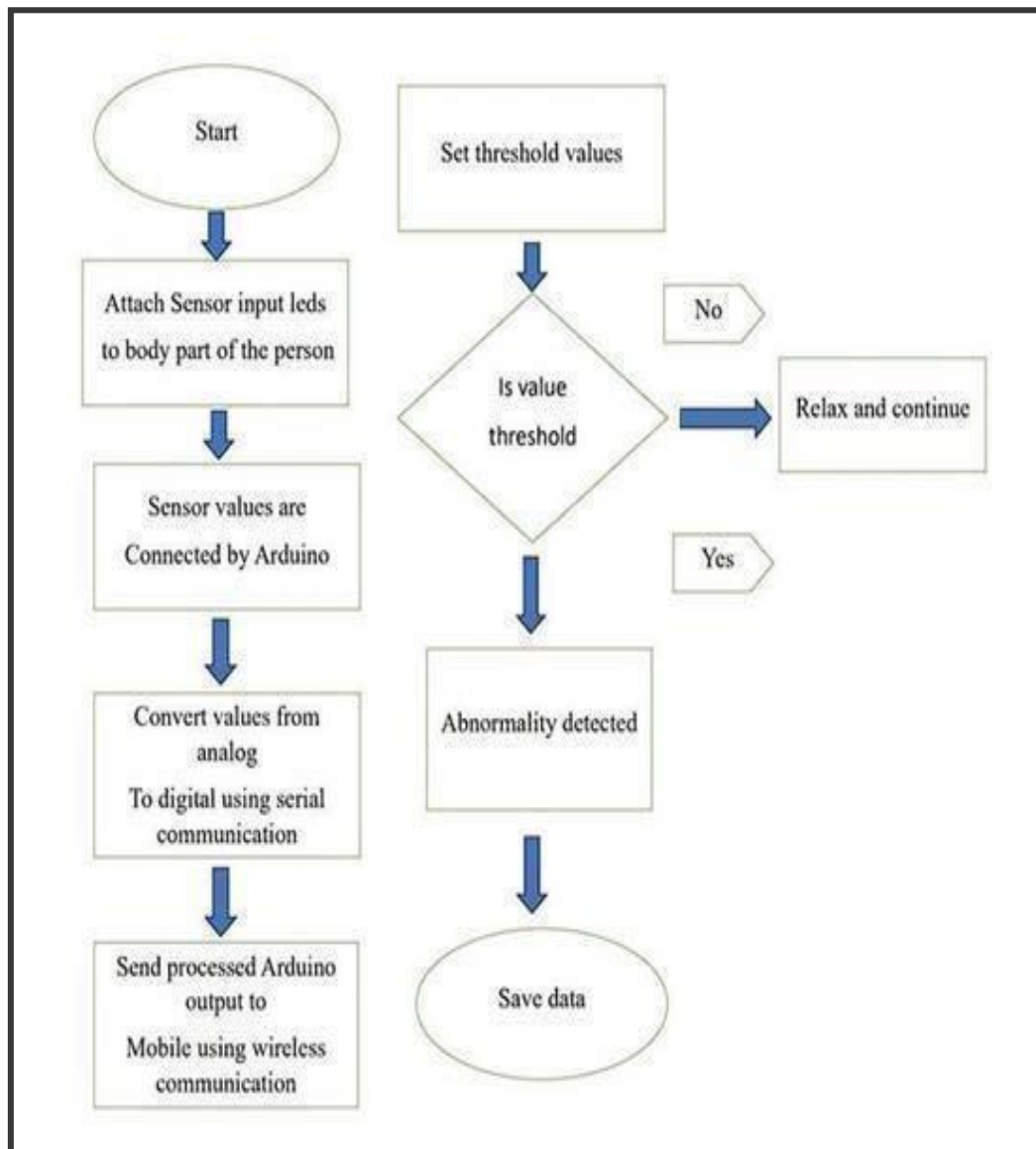
1. SERVING MECHANISM



2. HEALTH CARE MECHANISM



FLOW CHART



IV. WORKING

The "Elder Care Robot with Basic Healthcare and Serving Mechanism" is intended to offer senior citizens comprehensive assistance with everyday living and healthcare needs. Sophisticated software and hardware components work together to operate it. The healthcare role of the robot involves the realtime monitoring of vital indicators, including temperature and pulse, through the use of integrated sensors. A user-friendly interface is used to gather, process, and present this data to users, carers, and healthcare practitioners. The robot can initiate alerts, take preventive measures, or even offer aid in the case of anomalies or emergencies, such as notifying carers or contacting emergency services. Meals and drinks are served, and other everyday tasks are assisted by the robot's serving mechanism, which may consist of a robotic arm or other serving devices.

V. CONCLUSION

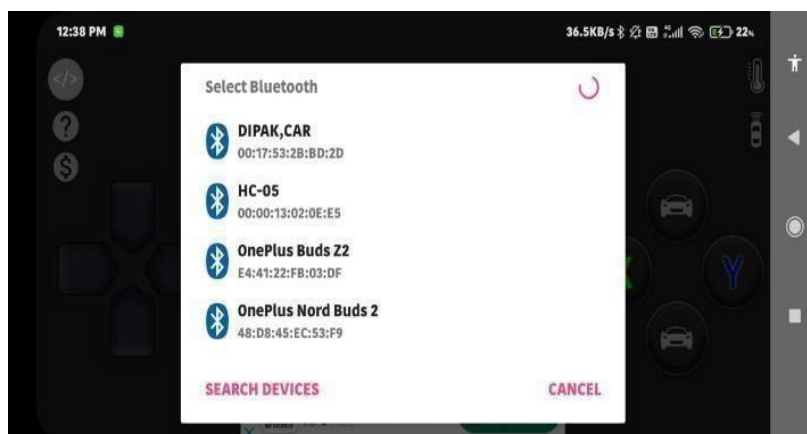
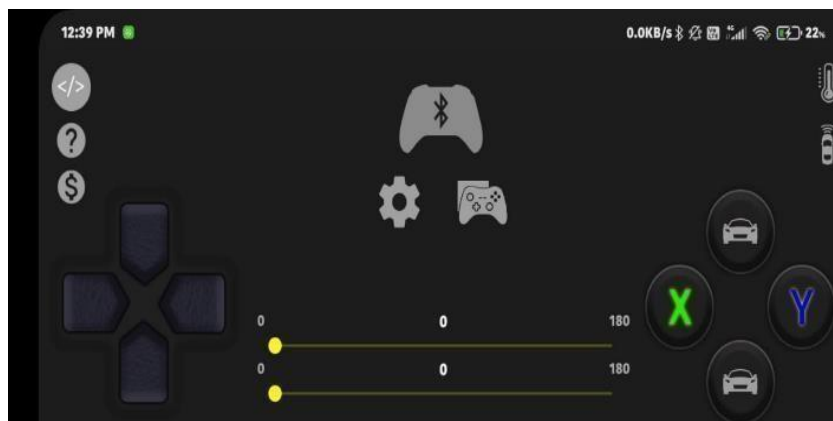
The "Elder Care Robot with Basic Healthcare and Serving Mechanism" represents a significant step forward in the field of elder care and healthcare support. With its capacity for real-time vital sign monitoring, user-friendly interface, and seamless integration of IoT technology, the robot effectively bridges the gap between healthcare and daily living assistance for elderly individuals.

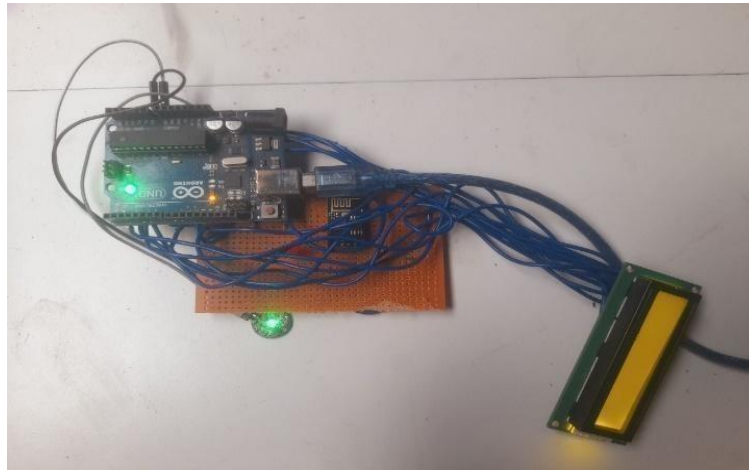
Its serving mechanism enhances the quality of life by promoting independence and comfort. The robot's safety features, including obstacle detection and emergency shutdown, provide a robust safety net, instilling confidence in users and caregivers. Moreover, the robot's commitment to data privacy and continuous improvement, guided by user feedback, further underscores its dedication to ensuring user wellbeing and satisfaction. This innovative solution not only enhances the quality of care for the elderly but also demonstrates the potential for technology to enrich the lives of those it serves

VI. RESULT

1. The robot provides real-time monitoring of vital signs, such as pulse and temperature, ensuring accurate and timely healthcare data collection
2. A user-friendly interface allows users and caregivers to easily access and interpret healthcare information and interact with the robot,
3. The robot successfully integrates IoT capabilities for remote monitoring and control, enhancing user safety and convenience
4. The robot effectively serves meals, drinks, and assists with daily tasks, promoting independence and comfort for elderly users
5. The robot incorporates obstacle detection mechanisms, ensuring safe navigation and reducing the risk of collisions
6. It features emergency shutdown options, adding an extra layer of safety and enabling immediate intervention when necessary
7. The robot complies with data privacy regulations and safeguards sensitive healthcare data to protect user privacy
8. An effective alert system is in place to notify caregivers and users of health anomalies or emergencies.
9. The robot's performance is continuously refined based on user feedback and iterative development, enhancing its overall effectiveness and user satisfaction.

Software Interface Display





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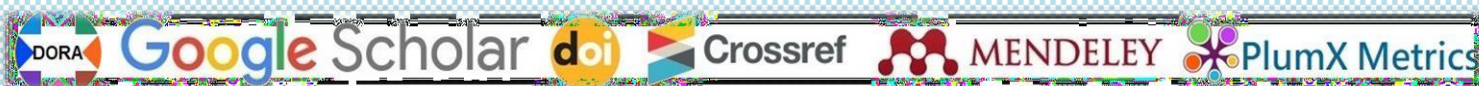
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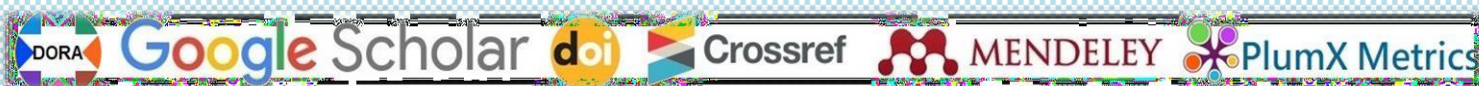
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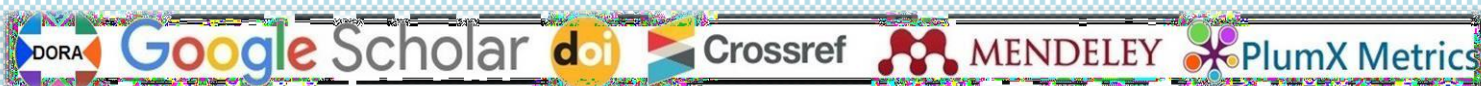
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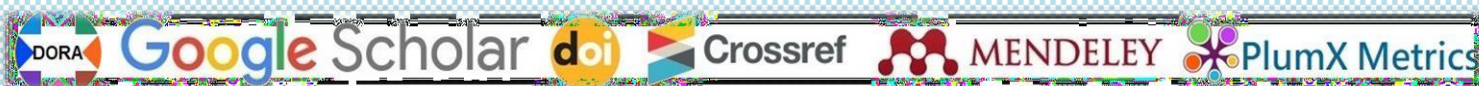
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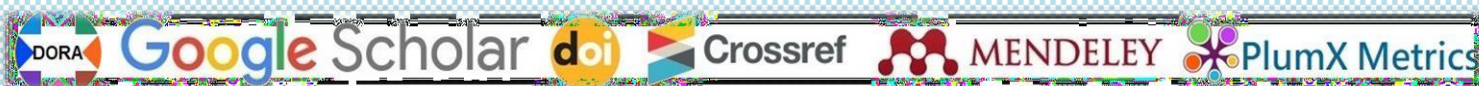
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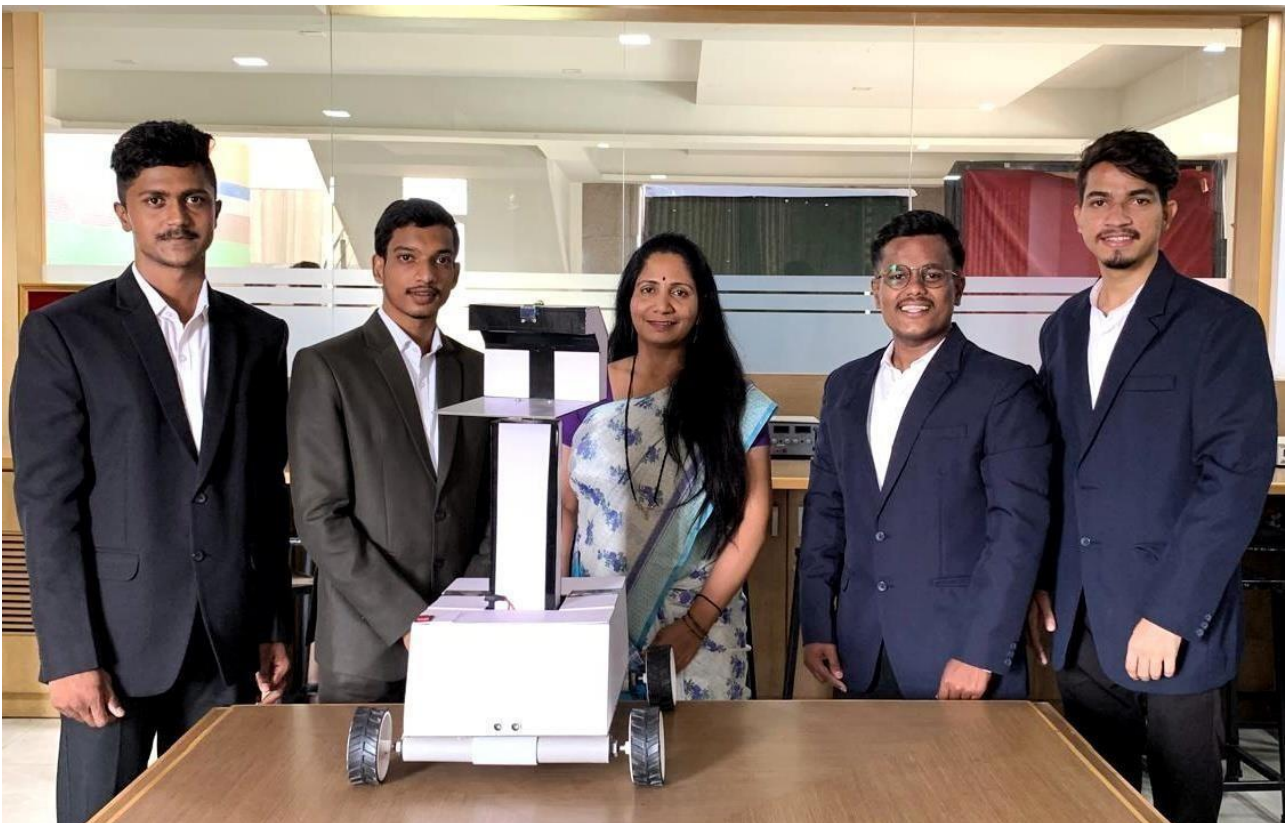
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